Enhanced geothermal systems: technology status, project economics, and the path to gigawatt scale

A cited research report

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Overview

Enhanced geothermal systems (EGS) generate geothermal electricity in settings that lack natural convective hydrothermal resources. Conventional geothermal power has been limited to places where heat, water, and rock permeability naturally coincide, yet most accessible geothermal energy resides in dry, impermeable rock [4]. EGS closes that gap by engineering permeability: stimulation techniques open pre-existing fractures or create new ones in low-permeability formations [1]. In the canonical scheme, high-pressure water is injected to trigger shear events that dilate existing cracks ("hydro-shearing"), water then circulates through the fracture network absorbing heat, and the returned hot fluid drives a steam turbine or binary plant before being reinjected [8]. Hydro-shearing is distinguished from the tensile hydraulic fracturing used in oil and gas, which also creates new fractures; notably, as long as injection pressure is maintained, proppants are not required to hold fractures open [8]. EGS plants are described as baseload resources, feasible in principle anywhere depending on resource depth, with favorable settings over deep granite beneath 3–5 km of insulating sediment and expected economic lifetimes of 20–30 years [8].

Technology status

The stimulation toolkit is broad: projects worldwide have combined hydraulic, chemical, thermal, carbon-based, and explosive methods, sometimes at the margins of existing hydrothermal fields where wells intersect hot but impermeable rock [2][7]. The global project record spans more than five decades and dozens of sites across Iceland, the USA, Germany, France, the UK, Japan, Australia, the Philippines, Mexico, Switzerland, Italy, and Indonesia, with documented stimulation methods for each [2][3]. The first deep, full-scale EGS reservoir was built at Fenton Hill, New Mexico (completed 1977 at ~2.6 km, 185 °C); it demonstrated that heat could be extracted at reasonable rates from hydraulically stimulated low-permeability hot crystalline rock, with a later 30-day flow test reaching ~190 °C production temperature and ~10 MW thermal, before budget cuts ended the work [14]. In Europe, the Soultz-sous-Forêts project connected a 1.5 MW demonstration plant to the grid and tested multiple stimulated zones and triplet (one injector / two producer) well configurations [18]. The largest EGS project cited is a 25 MW demonstration in Cooper Basin, Australia, a basin credited with 5,000–10,000 MW of potential [8][15].

The most consequential recent shift is the transfer of unconventional oil-and-gas practice into geothermal: long horizontal laterals and multi-stage hydraulic fracturing [1][2 (sq02)]. Recent and ongoing demonstrations — Fervo, Utah FORGE, and DOE EGS demonstrations — report successful creation of fracture networks in deep, hot, impermeable rock combined with long-term fluid circulation [1]. Supporting references include Fervo's use of oil drilling technology and a commercial-scale demonstration of a "first-of-a-kind" EGS system, FORGE as an international EGS technology laboratory, and reporting that Utah could host the world's largest EGS plant [20]. Complementary research directions include chemical stimulation (NLR–Utah FORGE work on concepts usable instead of, or alongside, mechanical stimulation) [1], downhole flow control and multi-stage horizontal well designs [10], multi-well and complex fracture-network configurations [12][15 (sq01)], and reservoir modeling and optimization using tools such as CMG STARS, TETRAD-G, TOUGH2, and slender-body-theory design optimization on high-performance computing [19][21].

Economics and policy

Cost benchmarks come mainly from two sources. The 2006 MIT/DOE assessment concluded EGS could produce electricity for as low as 3.9 cents/kWh, with costs sensitive to resource temperature, flow rate, drilling costs, and conversion efficiency [17]. Current policy targets are set by the DOE Geothermal Technologies Office's "Enhanced Geothermal Shot" (announced September 2022), aiming to cut EGS costs by 90% to $45/MWh by 2035 [23]. Note the tension: the 2006 figure (≈$39/MWh) is close to the 2035 target, implying the earlier estimate was aspirational rather than realized; the corpus does not reconcile these numbers directly. Projected levelized costs and underlying assumptions (capex, capacity factor, well productivity, decline, O&M) are compiled in the Electricity Annual Technology Baseline (2024) [24], and techno-economic tools SAM and GeoPHIRES are used to estimate site-based capital, O&M, and LCOE for hydrothermal and EGS projects [28].

On the resource side, MIT estimated U.S. EGS resources at 3–10 km depth exceeding 13,000 zettajoules, with over 200 ZJ extractable (potentially over 2,000 ZJ with better technology), recoverable capacity of 1.2–12.2 TW across conservative and moderate scenarios, and 100 GWe or more deployable by 2050 given a $1 billion, 15-year R&D program [17]. Public support instruments cited include 2009 DOE funding opportunities of up to $84 million over six years plus a $350 million ARRA solicitation with $80 million specifically for EGS [33], the Infrastructure Investment and Jobs Act's $84 million for four EGS demonstration projects, and the Inflation Reduction Act's extension of the production tax credit to 2024 and inclusion of geothermal in the new Clean Electricity PTC [32]. Outside the U.S., Geothermal Engineering Ltd received £15 million for deep geothermal in the UK, associated with the United Downs project in Cornwall [35].

Risks and open questions

Induced seismicity is the defining risk. Seismicity is common in EGS because of the high pressures involved, and events at The Geysers correlate with injection activity [34]. Induced seismicity in Basel led the city to suspend and then cancel its project [34], and the 2017 Mw 5.4 Pohang earthquake in South Korea may have been linked to the Pohang EGS project, which began in December 2010 targeting 1 MW; all research there stopped in 2018 [22][6]. The Australian government's position, as cited, is that hydrofracturing-induced seismicity risks are low relative to natural earthquakes and manageable through monitoring, and should not impede development — a view that sits in tension with the Basel and Pohang outcomes; the corpus presents both [34]. Mitigation practice includes traffic-light systems for deep well stimulation, as designed and implemented in Finland [6]. A cited synthesis of challenges across 64 EGS sites exists [27], though its findings are not summarized in the corpus.

Comparisons with alternatives are partial. Closed-loop geothermal (CLG) is attractive because heat production can be estimated with relatively high confidence, no reservoir stimulation is required (limiting induced seismicity risk and water use), and it can theoretically be applied anywhere — but faces complex downhole completions and difficulty creating sufficient heat-transfer area [29]. Superhot rock (>400 °C) could yield up to ten times the energy of a normal geothermal well but has not been harnessed for power due to significant technical challenges [30]. One cited study suggests EGS can be competitive with closed-loop/advanced and conventional hydrothermal systems under specific conditions [15], but the corpus provides no side-by-side cost figures.

Evidence gaps are worth flagging explicitly. The corpus contains no quantitative data on EGS drilling cost trends over time (only the existence of a 2025 geothermal drilling cost curve update [24]), no measured flow rates, well productivity, or thermal decline rates from the recent Fervo/FORGE demonstrations, no supply-chain or workforce quantification (constraints on skilled crews and high-temperature tools are asserted only in a loosely-supported reference entry [20 (sq14)]), and little on water use, land use, or permitting beyond the CLG comparison [29]. Statements about EGS deployment scale to date, or about generation currently on the grid, are not supported here beyond the Soultz 1.5 MW plant and Cooper Basin 25 MW demonstration [18][8].

What is the current technical state of the art in EGS reservoir

The defining problem: permeability, not heat

The technical core of enhanced geothermal systems is that the heat resource is abundant but the rock will not flow. Conventional geothermal power has historically been limited to places where "naturally occurring heat, water, and rock permeability are sufficient to allow energy extraction," whereas "most geothermal energy within reach of conventional techniques is in dry and impermeable rock" [4]. EGS therefore "requires augmentation of reservoir permeability using stimulation techniques that open pre-existing fractures or create new ones" [1], and the technologies "expand the availability of geothermal resources through stimulation methods, such as 'hydraulic stimulation'" [4]. Because the resource is not tied to natural convective plumbing, EGS is described as "apparently feasible anywhere in the world, depending on the resource depth," producing baseload power at a constant rate, with favourable settings typically over deep granite beneath a 3–5 km insulating sedimentary blanket, and with an expected plant economic lifetime of 20–30 years [8].

The stimulation toolkit is broader than "fracking"

The field has converged on a portfolio rather than a single method: "EGS projects have combined hydraulic, chemical, thermal, carbon-based, and explosive stimulation methods" [2][7]. Two important qualifications follow from the same source. First, a substantial share of what is catalogued as EGS is not greenfield hot-dry-rock: "Some EGS projects operate at the edges of hydrothermal sites where drilled wells intersect hot, yet impermeable, reservoir rocks. Stimulation methods enhance that permeability" [2][7]. Second, the mechanism most emphasised for EGS is *hydro-shearing* rather than tensile fracturing: pumping high-pressure water into naturally fractured rock raises fluid pressure and triggers "shear events that expand pre-existing cracks and enhance the site's permeability," and "as long as the injection pressure is maintained, high permeability is not required, nor are hydraulic fracturing proppants required to maintain the fractures in an open state" [8]. This is explicitly distinguished from the oil-and-gas hydraulic tensile fracturing that "can create new fractures in addition to expanding existing fractures" [8].

The historical project record documents each branch of the toolkit in the field:

- **Hydraulic**: the dominant method across the project inventory, from Fenton Hill (1973), Bad Urach and Falkenberg (1977), Le Mayet (1978), Hijiori (1985), Cooper Basin Habanero and Jolokia (2002), Basel (2006), Desert Peak and Brady Hot Springs (2008) onward [2]. Documented variants include high-volume, high-pressure injection with hydromechanical coupling evaluation at Le Mayet de Montagne [9]; "massive hydraulic fracturing in low permeable sedimentary rock" in the GeneSys project [13]; well-specific stimulation of well 27-15 at Desert Peak [13]; and hydraulic stimulation of the Upper Rhine Graben wells for power production [5].
- **Cyclic / pressure-cycling**: cyclic waterfrac stimulation has been advanced from "conceptual design and experimental results" for EGS development [11], and Salak (Indonesia) is listed as combining "chemical, thermal, hydraulic and cyclic pressure loading" [2].
- **Chemical**: Fenton Hill, Fjällbacka, Soultz, Groß Schönebeck, Coso, Paralana and Larderello are recorded as hydraulic *and* chemical; Altheim, Bacman, Mindanao, Leyte, Tiwi, Berlín, Unterhaching and Los Azufres as chemical [2]. Case histories include acid stimulation and scale/damage removal (calcium carbonate) in geothermal wells [11][5], coiled-tubing acid stimulation at Salak [5], acid treatment of well AZ-9AD at Los Azufres [5], acid stimulation of injectors at Leyte [11], and "enhanced permeability by chemical stimulation at the Berlín geothermal field" [11]. Chemical stimulation remains an active research front: a national laboratory is "partnering with Utah FORGE to test innovative chemical stimulation concepts to enhance reservoir permeability," intended for use "instead of, or in tandem with, mechanical stimulation methods" [1]. A known limitation is that "fines migration poses challenge for reservoir-wide chemical stimulation of geothermal carbonate reservoirs" [10].
- **Thermal**: Krafla, Sumikawa, Bouillante and Hellisheidi are listed as thermal, Mosfellssveit as thermal and hydraulic [2]; combined "hydraulic and thermal stimulation" was executed at Raft River, Idaho, as a DOE EGS project [3][6].
- **Explosive**: Rosemanowes (hydraulic and explosive) and Geysers Unocal (explosive) [2].
- **Mechanical/near-well access**: radial jetting technology was tested to enhance the geothermal system at the Klaipėda demonstration plant [6].

Multi-stage horizontal wells, zonal isolation and flow control

The most consequential recent shift is the wholesale importation of unconventional oil-and-gas well architecture. “Recent advancements include applying oil and gas developed techniques such as long horizontal laterals and multi-stage hydraulic fracturing to geothermal settings” [1], and Fervo has been reported as hitting a “milestone in using oil drilling technology to tap geothermal energy” [20], alongside a preprint reporting a “Commercial-Scale Demonstration of a First-of-a-Kind Enhanced Geothermal System” [20].

Beyond the fracturing itself, the engineering frontier is controlling *where* fluid enters and exits a long lateral. The literature identifies downhole flow control as “a key for developing enhanced geothermal systems in horizontal wells” [10], with companion work on “real-time temperature monitoring and flow control for improved heat extraction in enhanced geothermal systems” [12]. Well-architecture optimisation is being explored numerically rather than only empirically: studies address well layout schemes and fracture parameters (including water-cooling effects) on heat production [10][12], multi-well injection configurations [15][12], multifracture systems with matrix permeability enhancement from thermal unloading [10], and spiral wellbore geometries in horizontal wells for hot-dry-rock heat extraction (Gonghe Basin, Qinghai) [12].

A caveat on zonal isolation specifically. Several corpus entries assert that EGS projects are “advancing technologies such as multi-stage horizontal wells and zonal isolation to enhance reservoir performance” [6] and that laboratory modelling supports “multi-stage horizontal wells and zonal isolation” designs [19]. However, the underlying material in those entries is bibliographic (reference lists) and general modelling-capability description; the corpus contains **no field data on isolation hardware, packer/plug performance at temperature, stage counts, or per-stage flow allocation**. That specific claim should therefore be treated as weakly evidenced here.

Demonstrated field performance

The corpus supports the following as the concrete, quantified performance record:

- **Fenton Hill, New Mexico (first deep, full-scale EGS reservoir)**: reservoir completed 1977 at ~2.6 km depth in 185 °C rock; enlarged in 1979 by additional hydraulic stimulation and operated about one year, demonstrating “that heat could be extracted at reasonable rates from a hydraulically stimulated region of low-permeability hot crystalline rock.” A second reservoir underwent a 30-day flow test with 20 °C constant reinjection temperature; production temperature rose steadily to ~190 °C, corresponding to a thermal power level of about 10 MW. Budget cuts ended the work [14].
- **Soultz-sous-Forêts, France**: a 1.5 MW demonstration plant connected to the grid; the project “explored the connection of multiple stimulated zones and the performance of triplet well configurations (1 injector/2 producers)” [18] — i.e., the first systematic field test of multi-zone connectivity and multi-producer geometry.
- **Cooper Basin, Australia**: described as “the world’s largest EGS project,” a 25-megawatt demonstration plant, in a basin with stated potential of 5,000–10,000 MW [8].
- **Recent demonstrations**: “Recent and ongoing demonstrations (e.g., Fervo, FORGE, DOE EGS demonstrations) show successful creation of a fracture network in deep and hot impermeable rocks combined with long-term fluid circulation” [1]. Utah FORGE is characterised as “an international laboratory for enhanced geothermal system technology development” [6][20], and press coverage indicates Utah “could become home to world’s largest enhanced geothermal plant” [20].
- **Other documented demonstrations without quantified output in this corpus**: the Northwest Geysers EGS Demonstration Project (characterisation and reservoir response to injection) [3]; Newberry Volcano EGS Demonstration results 2010–2014 [3]; Raft River hydraulic and thermal stimulation [3][6]; the South Hungary EGS (SHEGS) Demonstration Project [6]; Reykjanes DEEPEGS well IDDP-2 [20]; United Downs Deep Geothermal Power Project [20]; Qiabuqia (Northwest China) EGS power generation research [20].
- **Failures and terminations are part of the state of the art.** Basel: “Induced seismicity in Basel led to the cancellation of the EGS project there” [18], with the reservoir characterised in the literature [13]. Pohang, South Korea: started December 2010 with a 1 MW goal; “the 2017 Pohang earthquake may have been linked to the activity of the Pohang EGS project. All research activities were stopped in 2018” [22][6]. More generally, “seismicity is common in EGS, because of the high pressures involved,” and induced seismicity “varies from

site to site and should be assessed before large scale fluid injection" [34]. Risk-management practice now includes engineered traffic-light systems for deep geothermal well stimulation, as designed and implemented in Finland [6]. A synthesis of "challenges in developing enhanced geothermal systems (EGS): observations from 64 EGS sites" exists in the literature [27], indicating that difficulties are systemic rather than isolated.

Conflict/dating note: [8] states the largest EGS project is the 25 MW Cooper Basin demonstration, while [20] reports Utah as a candidate for the "world's largest enhanced geothermal plant" (retrieved 2024) and the Cooper Basin work is discussed retrospectively ("What did we learn about EGS in the Cooper Basin?") [11]. These statements are plausibly of different vintages; the corpus does not resolve which is currently accurate.

Characterisation, modelling and design optimisation

Reservoir creation is now paired with subsurface characterisation and simulation as a co-equal capability. The national laboratory position is that work is "focusing on characterizing the subsurface to ensure accurate, onsite understanding of EGS to allow for optimization of fracture networks and strategic development and operational decisions," with the other key area being "drilling analysis and techno-economic modeling to optimize deep drilling efficiencies, a critical step in developing EGS" [1]. Practically, this involves 3D static and dynamic simulation "including thermal, hydrological, mechanical, and chemical properties," run on high-performance computing "from well targeting to reservoir operations and from optimization to design of unconventional geothermal systems" [19], using tools such as CMG STARS, TETRAD-G and the TOUGH2 suite, plus in-house slender-body-theory reservoir design optimisation models that screen "large numbers of reservoir designs and well configurations while minimizing computational run time" [19][21]. Supporting method development includes embedded 3D fracture modelling for fracture-dominated flow and heat transfer [21] and code modifications for chemical tracers and embedded natural fractures at the EGS Collab [21]. Techno-economic coupling is provided by open-source tools SAM and GEOPHIRES, which estimate site-based capex, O&M and LCOE for hydrothermal or EGS technologies and model reservoir performance and heat extraction over time [28]. Application examples include an EGS reservoir model and techno-economic simulation for campus heating with Cornell University [1], and resource characterisation for temperature-at-depth and EGS potential mapping [31][26].

The academic simulation literature is converging on fully coupled thermal–hydraulic–mechanical(–chemical) representations of fracture networks: extended embedded discrete fracture models capturing geometrical and topological complexity [16], coupled THM analysis of stimulation and production [16], integrated THMC plus wellbore-dynamics simulation of fracture-network roles [15][25], PD-HT-FEM simulation of fracture networks and heat extraction [25], rough/heterogeneous fracture network effects [15], natural-fracture effects on heat extraction under THM [25], multi-scale fully coupled lifecycle optimisation [10], and stimulation-strategy differentiation between tight rock and naturally fractured formations (Songliao Basin) [16]. Boundary-condition choices are known to materially affect performance predictions [12]. A recent review of "reservoir stimulation technologies for enhanced geothermal systems" consolidates this field [10].

Drilling technology evolution and the cost question

What the evidence supports. Drilling capability is stated in terms of depth and temperature reach: "advanced drilling techniques can penetrate hard crystalline rock to depths of up to, or exceeding, 15 km, providing access to higher-temperature rock (400 °C and above) as temperature increases with depth" [8]. The direction of technology flow is explicitly from oil and gas to geothermal: long horizontal laterals and multi-stage hydraulic fracturing are "oil and gas developed techniques" being applied "to geothermal settings" [1], and Fervo's reported milestone is framed precisely as "using oil drilling technology to tap geothermal energy" [20]. Drilling cost is recognised as one of four principal sensitivities for EGS electricity cost, alongside resource temperature, fluid flow through the system, and power conversion efficiency [17].

Cost targets rather than measured cost curves. The 2006 MIT/DOE assessment claimed EGS "could produce electricity for as low as 3.9 cents/kWh" and estimated that with $1 billion of R&D over 15 years, 100 GWe or more could be available in the United States by 2050, with recoverable resources of 1.2–12.2 TW [17]. The current policy target is the Enhanced Geothermal Shot announced in September 2022, aiming "to reduce the cost of EGS by 90%, to $45/megawatt hour by 2035" [23]. Cost-tracking infrastructure exists — the Electricity Annual Technology Baseline (2024) "includes projected levelized costs of electricity for enhanced geothermal systems (EGS), along with key cost and performance assumptions such as capex, capacity factor, well productivity, decline, and O&M" [24] — and there is dedicated work on "Geothermal Drilling Cost Curve Updates" presented at the Stanford Geothermal Workshop (2025) [24], plus a Geothermal Drilling Program historically [35] and "Enhanced Geothermal Shot" technical reporting [24].

Where the evidence is missing. The corpus does **not** contain:

1. Any numerical drilling cost figures, cost-per-metre or cost-per-well time series for geothermal wells, nor the contents of the drilling cost curve update [24] — only the existence of that work.
2. Any decomposition of observed cost reduction into (a) oil-and-gas technology transfer versus (b) learning-by-doing / experience-curve effects. The corpus supports only the qualitative statement that O&G techniques have been transferred [1][20]; it provides no attribution shares, no learning rates, and no well-to-well cost decline data. Any quantitative split would be invented.
3. Quantified field performance metrics for the recent horizontal multi-stage demonstrations — flow rates, wellhead temperatures, productivity/injectivity indices, per-stage contributions, thermal drawdown, or power output for Fervo/FORGE/DOE demonstrations. The corpus asserts only qualitative success: fracture-network creation in deep hot impermeable rock plus long-term fluid circulation [1], and a commercial-scale demonstration preprint whose results are not reproduced here [20].
4. Field evidence on zonal-isolation hardware performance at EGS temperatures (see caveat above) [6][19].
5. Data on crew, rig or high-temperature tool supply constraints at scale; one corpus entry gestures at "the need for specialized skills and advanced tools, such as high-temperature equipment" [20], but the underlying text is a reference list and does not substantiate a quantified constraint. Relatedly, superhot rock (>400 °C) systems, which would have up to ten times the energy output of a normal geothermal well, "have not yet been harnessed for power production due to significant technical and technological challenges," including advanced drilling methods, materials and reservoir monitoring [30].

Synthesis

The demonstrated technical state of the art can be summarised as follows. Stimulation is a mature, diversified practice with five decades of field trials across hydraulic, hydro-shear, cyclic, chemical, thermal and (historically) explosive methods [

What is the global and US pipeline of EGS and advanced geothermal

The documented project inventory is long, but capacity- and developer-level data are largely absent from the corpus

The most comprehensive pipeline listing in the evidence base is the project table maintained in the Wikipedia EGS article, which enumerates EGS and EGS-adjacent stimulation projects "around the world" with name, country, state/region, start year, and stimulation method [2]. That table documents a continuous global pipeline stretching back more than five decades: Mosfellssveit, Iceland (1970, thermal and hydraulic); Fenton Hill, USA–New Mexico (1973, hydraulic and chemical); Bad Urach and Falkenberg, Germany (1977); Rosemanowes, UK (1977, hydraulic and explosive); Le Mayet, France (1978); East Mesa, California (1980); Krafla, Iceland (1980); Baca, New Mexico (1981); Geysers Unocal, California (1981, explosive); Beowawe, Nevada (1983); Fjällbacka, Sweden (1984); Hijiori, Japan (1985); Soultz, France (1986); Ogachi and Sumikawa, Japan (1989); Tyrnyauz, Russia (1991); Groß Schönebeck, Germany (2000); Cooper Basin: Habanero and Jolokia 1, Australia (2002); Coso, California (1993, 2005); Paralana and Olympic Dam, Australia (2005); Basel, Switzerland (2006); Larderello, Italy (1983, 2006); Insheim, Germany (2007); Desert Peak, Nevada (2008); Brady Hot Springs, Nevada (2008); Southeast Geysers, California (2008); and Genesys: Hannover, Germany (2009), with the excerpt truncating mid-entry at "St." (i.e., St. Gallen) [2]. Additional projects appear in the reference apparatus rather than the table: Newberry Volcano EGS Demonstration (2010–2014) [3], the Northwest Geysers EGS Demonstration Project in California [3], Raft River, Idaho (a DOE EGS hydraulic and thermal stimulation program) [3], Soda Lake, Nevada [3], Mauerstetten in the Bavarian Molasse Basin [3], Rittershoffen (wells GRT-1/GRT-2), France [6], the Klaipėda radial-jetting demonstration in Lithuania [6], a deep geothermal stimulation with a traffic-light system in Finland [6], the South Hungary Enhanced Geothermal System (SHEGS) Demonstration Project [6], Utah FORGE [6][20], the Reykjanes DEEPEGS demonstration well IDDP-2 [20], petrothermal generation in crystalline rocks in Germany [20], the United Downs Deep Geothermal Power Project in the UK [20], the Eden Project [20], and the Qiabuqia EGS power-generation project in northwest China [20]. A separate synthesis in the same reference list reports observations from "64 EGS sites," indicating the global historical population of EGS sites is on that order [27].

Two important caveats apply to reading this list as a "pipeline." First, many entries are chemical or thermal well-stimulation campaigns at existing hydrothermal fields rather than greenfield hot-dry-rock reservoirs: the source explicitly notes that "some EGS projects operate at the edges of hydrothermal sites where drilled wells intersect hot, yet impermeable, reservoir rocks," with stimulation used to enhance permeability [2][7]. Second, the table carries no capacity, developer, or operating-status columns [2] — so the corpus does not support a project-by-project MW tally, ownership attribution, or commissioning schedule.

Operating, demonstrated, and terminated: what the evidence does establish about status

- **Cooper Basin, Australia** is described as the world's largest EGS project, a 25-megawatt demonstration plant, at a resource with stated potential to generate 5,000–10,000 MW [8][15 n/a]. The same source states EGS systems are under development in Australia, France, Germany, Japan, Switzerland, and the United States [8].
- **Soultz-sous-Forêts, France** (EU R&D project, Alsace) connects a **1.5 MW demonstration plant to the grid**, and explored connection of multiple stimulated zones and triplet well configurations (1 injector / 2 producers) [18].
- **Basel, Switzerland** was suspended and then cancelled because of induced seismicity [18][34].
- **Pohang, South Korea** began in December 2010 with a 1 MW target; the 2017 Pohang earthquake may have been linked to project activity and all research activities were stopped in 2018 [22].
- **Fenton Hill, USA** (Los Alamos) was the first attempt at a deep, full-scale EGS reservoir: reservoir completed 1977 at ~2.6 km and 185 °C, enlarged with additional stimulation in 1979 and operated about a year; a second reservoir gave a 30-day flow test reaching ~190 °C production temperature, ~10 MW thermal, before budget cuts ended the study [14].
- **Fervo Energy** achieved a milestone using oil-and-gas drilling technology for geothermal, reported in July 2023, alongside a preprint describing a "Commercial-Scale Demonstration of a First-of-a-Kind Enhanced Geothermal System" [20]. A cited news item indicates Utah "could become home to world's largest enhanced geothermal plant" [20], and Utah FORGE is characterized as an international laboratory for EGS technology development [6][20].
- **Recent demonstrations collectively (Fervo, FORGE, DOE EGS demonstrations)** are reported to show successful creation of fracture networks in deep, hot, impermeable rock combined with long-term fluid circulation, enabled by long horizontal laterals and multi-stage hydraulic fracturing [1].
- **US federal demonstration pipeline:** the Infrastructure Investment and Jobs Act authorized $84 million to support EGS development through **four demonstration projects** [32]; earlier DOE funding opportunity announcements in 2009 offered up to $84 million over six years, plus a $350 million ARRA solicitation including $80 million specifically for EGS [33].
- **United Downs (Cornwall, UK)** appears with a project update through the 2022 European Geothermal Congress, a £15 million funding award to GEL for UK deep geothermal in 2023, and a BBC item titled "Earth's heat to produce electricity for homes in UK clean energy first" dated 2026 [20][35].

- **Direct-use/campus pipeline:** NLR is partnering with Cornell University to develop EGS reservoir models and techno-economic simulations for campus heating [1], consistent with Cornell's Earth Source Heat program and a $7.2M exploratory research grant referenced in the corpus [35].

Advanced (non-EGS) geothermal. For closed-loop geothermal (CLG), the corpus describes technology attributes and NLR's R&D role — one or more wells with fluid circulating in a closed loop, no reservoir stimulation required (limiting induced-seismicity risk and water use), theoretically deployable anywhere, with challenges in complex downhole completions and creating sufficient heat-transfer area [29] — but names **no CLG projects, developers, capacities, or timelines**. Superhot rock (>400 °C supercritical) systems, with estimated output up to 10× a normal geothermal well, "have not yet been harnessed for power production due to significant technical and technological challenges" [30]. NLR modeling spans traditional hydrothermal sites, EGS, and closed-loop advanced systems [19], including thermal and well-flow performance of closed-loop geothermal in the Wattenberg area [24].

Evidence gap: the corpus contains no named-project pipeline table with developer, nameplate MW, location, and commercial-operation dates for the current generation of US or global EGS/AGS projects (e.g., Fervo's project portfolio capacities, FORGE's planned plant size, or the four IIJA demonstration sites). Any specific capacity or date beyond those cited above would be invention.

Levelized cost of electricity for EGS: two anchors, and a tension between them

The corpus provides two quantitative cost anchors and they are difficult to reconcile:

1. **MIT/DOE (2006):** "EGS could produce electricity for as low as 3.9 cents/kWh" (~$39/MWh), with costs identified as sensitive to four main factors — resource temperature, fluid flow through the system, drilling costs, and power conversion efficiency [17]. The same report put US EGS resources at 3–10 km depth above 13,000 ZJ (over 200 ZJ extractable, potentially over 2,000 ZJ with better technology), recoverable resources of 1.2–12.2 TW under conservative and moderate scenarios, and 100 GWe or more available by 2050 in the US with a $1 billion, 15-year R&D investment [17].
2. **DOE Enhanced Geothermal Shot (September 2022):** the goal is to **reduce the cost of EGS by 90%, to $45/MWh by 2035** [23][32-n/a]. Taken at face value, a 90% reduction target implies a starting point on the order of $450/MWh — roughly an order of magnitude above the 2006 MIT "as low as 3.9 cents/kWh" figure [17][23].

The corpus does not explain this discrepancy. The most defensible reading is that the MIT number was a forward-looking, best-case projection contingent on successful R&D [17], while the 2022 Earthshot baseline reflects observed costs of then-existing EGS; but the evidence base does not state the Earthshot baseline value explicitly, so this remains an inference from the "90% / $45 per MWh" framing [23].

Cost and performance assumptions behind EGS LCOE: the corpus identifies the parameter set but not the numbers

The evidence explicitly confirms that a current, authoritative parameter set exists: the **Electricity Annual Technology Baseline (2024)** "includes projected levelized costs of electricity for enhanced geothermal systems (EGS), along with key cost and performance assumptions such as capex, capacity factor, well productivity, decline, and O&M" [24]. However, **no numeric values for capex ($/kW), capacity factor (%), per-well productivity (flow rate), thermal decline rate (%/yr), or O&M ($/kW-yr or $/MWh) appear anywhere in the corpus.** This is the single largest quantitative gap for this section, and it should be filled from the ATB itself rather than estimated.

What the corpus does supply is the qualitative structure of the cost model and the tooling used to build it:

- **Modeling tools.** NLR develops open-source techno-economic tools — the **System Advisor Model (SAM)**, which "enables users to estimate site-based capital costs, operations and maintenance costs, and levelized costs of electricity of hydrothermal or EGS technologies," with hourly performance simulation, financial

models, and policy impacts; and **GEOPHIRES**, which offers dynamic modeling of hydrothermal and EGS electricity plus direct-use applications "with a strong focus on reservoir performance and heat extraction over time" [28]. Outputs from this work inform geospatial analysis, capacity-expansion models, and next-generation technology cost targets [28]. Related products include "Geothermal Drilling Cost Curve Updates" (2025), a Geothermal Module in reV, an Enhanced Geothermal Shot technical report (2023), and GEOPHIRES-X applications beyond electricity [24].
- **Drilling cost as a first-order driver.** Drilling costs are one of MIT's four LCOE sensitivities [17]; NLR emphasizes "drilling analysis and techno-economic modeling to optimize deep drilling efficiencies, a critical step in developing EGS" [1]; and Fervo's transfer of oil-and-gas drilling technology is presented as the key recent lever [20][1].
- **Plant life and dispatch profile.** EGS plants are expected to have an economic lifetime of **20–30 years** [8] and are baseload resources producing power at a constant rate [8] — the qualitative basis for a high assumed capacity factor, though no percentage is given in the corpus.
- **Resource/temperature assumptions.** Good sites are typically over deep granite beneath 3–5 km of insulating sediment [8]; advanced drilling can reach hard crystalline rock at up to or beyond 15 km, accessing rock at 400 °C and above [8]; and higher-enthalpy supercritical conditions (>400 °C) could yield up to 10× the output of a normal well, materially improving project economics if the technical barriers are solved [30].
- **Well productivity and decline as design variables.** The literature cited treats per-well flow and heat-extraction performance as engineering outcomes of well and fracture design: multi-stage horizontal wells with downhole flow control [10], real-time temperature monitoring and flow control [12], well-layout and fracture-parameter optimization including water-cooling effects [10], multi-well injection configurations [15], spiral horizontal wellbores [12], and natural-fracture-network effects on heat extraction under coupled thermal-hydraulic-mechanical(-chemical) conditions [25]. Reservoir permeability and porosity are shown to govern development-model performance [15]. Integrated techno-economic comparisons have also been published for EGS coupled with carbon storage, indicating conditions under which EGS can be competitive [15]. None of these citations provide the specific flow rates or decline percentages used in ATB-style LCOE calculations.
- **Policy inputs that lower effective cost.** The Inflation Reduction Act extended the renewable production tax credit (including geothermal) through 2024 and included geothermal in the new Clean Electricity PTC beginning in 2024 [32]; SAM explicitly incorporates policy impacts into LCOE estimates [28].

Bottom line

The corpus supports a robust qualitative picture: a five-decade, multi-continent pipeline of roughly several dozen EGS and stimulation projects [2][27], of which the demonstrably grid-connected or largest historical entries are Soultz (1.5 MW) [18] and Cooper Basin (25 MW demonstration) [8], with notable terminations at Basel [18], Pohang [22], and Fenton Hill [14]; a current wave centered on Fervo, Utah FORGE, and four DOE/IIJA demonstration projects in the US [1][20][32], plus United Downs in the UK [20][35], Reykjanes/DEEPEGS in Iceland [20], Qiabuqia in China [20], and campus direct-use work at Cornell [1][35]; and an early-stage, project-unnamed advanced geothermal segment in closed-loop [29] and superhot-rock [30] configurations. On cost, the corpus gives only two anchor numbers — 3.9 ¢/kWh as a 2006 aspiration [17] and $45/MWh by 2035 as a 2022 policy target representing a 90% reduction [23] — and confirms that the detailed capex/capacity-factor/well-productivity/decline/O&M assumption set resides in the 2024 Electricity ATB and in SAM/GEOPHIRES [24][28] without reproducing any of those values. Filling in the numeric assumption table, current-project capacities, and commercial-operation dates requires sources beyond this corpus.

How does EGS compare economically and technically with

What each approach requires physically. Conventional hydrothermal power requires the natural co-occurrence of heat, water, and rock permeability sufficient to allow energy extraction, which restricts it to a limited set of sites [4]. EGS removes the permeability constraint by engineering it: because most geothermal energy within reach of conventional techniques sits in dry and impermeable rock, EGS uses stimulation methods — principally hydraulic stimulation — to expand the accessible resource [4]. The canonical EGS mechanism is hydro-shearing: pumping high-pressure water down an injection well into naturally fractured rock raises fluid pressure, triggering shear events that expand pre-existing cracks and increase permeability; as long as injection pressure is maintained, neither high native permeability nor proppants are needed to hold fractures open [8]. This is distinguished in the corpus from hydraulic tensile fracturing as used in oil and gas, which can create wholly new fractures in addition to expanding existing ones [8]. In practice EGS projects have used a wider palette than hydraulics alone — hydraulic, chemical, thermal, carbon-based, and explosive stimulation — and some operate at the margins of existing hydrothermal fields where wells intersect hot but impermeable rock [2][7]. Historical project records show this diversity: Fenton Hill (hydraulic and chemical), Rosemanowes (hydraulic and explosive), Krafla and Sumikawa (thermal), Altheim and several Philippine fields (chemical), and Salak (chemical, thermal, hydraulic and cyclic pressure loading) [2].

Closed-loop/advanced geothermal (CLG/AGS) is defined in the corpus as one or more wells drilled into hot rock with fluid circulating through a closed-loop system to bring heat to surface for direct heating and electricity, with multiple proposed designs based on different geometries and heat transfer fluids [29]. The pivotal technical difference is explicit: EGS "requires augmentation of reservoir permeability using stimulation techniques that open pre-existing fractures or create new ones," *unlike* advanced geothermal systems [1]; correspondingly, in CLG "reservoir stimulation is not required" [29].

Claimed advantages of each.

| Attribute | Evidence for EGS | Evidence for CLG/AGS | Evidence for hydrothermal | |—|—|—|—| | Siting breadth | "apparently feasible anywhere in the world, depending on the resource depth" [8] | "can theoretically be applied anywhere" [29] | limited to sites with natural heat + water + permeability [4] | | Output character | baseload resource producing power at a constant rate [8] | designs targeted at direct heating and electricity [29] | — (not characterised in corpus) | | Performance predictability | requires subsurface characterisation to optimise fracture networks and development decisions [1] | "heat production can be estimated with relatively high confidence" [29] | — | | Stimulation-linked risk | seismicity "common in EGS, because of the high pressures involved" [34] | no stimulation → "limits the risk of induced seismicity and lowers water consumption" [29] | — | | Engineering upside | oil-and-gas transfer: long horizontal laterals, multi-stage hydraulic fracturing [1] | "engineered system… opportunities for optimization and scale-up with technological improvements" [29] | — |

Both novel reservoir designs are framed by NLR as potentially reducing upfront project risk and enabling geothermal development "anywhere in the United States," relative to traditional hydrothermal sites [19].

Demonstrated technical performance. EGS has a fifty-year experimental record. Fenton Hill, the first deep full-scale EGS reservoir, was completed in 1977 at ~2.6 km in 185 °C rock; after enlargement by additional hydraulic stimulation it operated about a year and demonstrated that heat could be extracted at reasonable rates from a hydraulically stimulated region of low-permeability hot crystalline rock, and a 1986 second reservoir achieved ~190 °C production temperature and ~10 MW thermal in a 30-day flow test before budget cuts ended the work [14]. Soultz-sous-Forêts connected a 1.5 MW demonstration plant to the grid and tested connection of multiple stimulated zones and triplet (1 injector/2 producer) well configurations [18]. The largest EGS project cited is a 25 MW demonstration plant at Cooper Basin, Australia [8]. Most recently, demonstrations including Fervo, Utah FORGE, and DOE EGS projects show successful creation of fracture networks in deep, hot, impermeable rock combined with long-term fluid circulation [1], with Fervo reported as reaching a milestone in applying oil-drilling technology to geothermal and as a "commercial-scale demonstration of a first-of-a-kind enhanced geothermal system" [20]. By contrast, the corpus contains no comparable record of demonstrated CLG field performance — CLG is discussed only as designs, attractions, and challenges [29], with NLR modelling work such as thermal and well flow performance of closed-loop geothermal in the Wattenberg area [24]. This asymmetry in demonstration maturity is itself an important comparative finding.

Economic comparison

The corpus provides only fragmentary and partly contradictory cost figures, and no like-for-like EGS-vs-CLG-vs-hydrothermal LCOE comparison.

- The 2006 MIT/DOE report claimed EGS "could produce electricity for as low as 3.9 cents/kWh" (~$39/MWh), and identified four dominant cost sensitivities: resource temperature, fluid flow through the system, drilling costs, and power conversion efficiency [17].
- In September 2022 DOE's Geothermal Technologies Office announced an "Enhanced Geothermal Shot" aiming to cut EGS cost by 90%, to $45/MWh by 2035 [23][32-adjacent context in 23].

These two statements are in tension: MIT's 2006 figure of ~$39/MWh is *below* the level DOE in 2022 characterised as requiring a 90% cost reduction to reach by 2035 [17][23]. The most defensible reading is that MIT's number was a technology-potential projection contingent on R&D success, whereas the Earthshot baseline reflects actual near-term costs an order of magnitude higher; the corpus does not state this reconciliation explicitly, so it should be treated as an unresolved conflict rather than a settled fact.

Institutional cost machinery exists but the numbers are not in the corpus. The System Advisor Model (SAM) lets users estimate site-based capital costs, O&M costs, and LCOE for hydrothermal *or* EGS technologies, with financial models, hourly performance simulation and policy impacts; GeoPHIRES provides dynamic modelling of hydrothermal and EGS electricity and direct-use applications with a focus on reservoir performance and heat extraction over time [28]. The Electricity Annual Technology Baseline (2024) includes projected LCOEs for EGS together with capex, capacity factor, well productivity, decline and O&M assumptions [24]. Two economic parameters relevant to any comparison do appear: EGS plants are expected to have an economic lifetime of 20–30 years [8], and EGS is a baseload (constant-output) resource [8], which matters for value per MWh but is not monetised anywhere in the corpus.

One corpus item asserts, on the basis of an integrated techno-economic feasibility study of EGS with carbon storage, that EGS "can be competitive with closed-loop/advanced geothermal (AGS) and conventional hydrothermal systems under specific conditions" [15]. This should be flagged as weak: the underlying evidence is a bibliographic title (Xue et al., *Applied Energy*, 2024) rather than reported results, and the comparison conditions are unstated [15]. A similar caution applies to a corpus entry asserting that EGS's technical and economic challenges are "comparable to those of closed-loop/advanced geothermal (AGS) and conventional hydrothermal systems, though they differ in implementation and resource requirements" [27] — the supporting text is a reference list, not an analysis, so the claim is essentially unsourced within the corpus.

Public support has been a material part of EGS economics: DOE issued two 2009 funding opportunity announcements offering up to $84 million over six years, plus a $350 million Recovery Act announcement including $80 million specifically for EGS [33]; the Infrastructure Investment and Jobs Act authorised $84 million for four EGS demonstration projects; and the Inflation Reduction Act extended the renewable production tax credit (including geothermal) to 2024 and included geothermal in the new Clean Electricity PTC from 2024 [32]. No equivalent CLG or hydrothermal support figures are given, so relative subsidy intensity cannot be assessed.

Credible criticisms of each approach

EGS.

1. *Induced seismicity is intrinsic, not incidental.* "Seismicity is common in EGS, because of the high pressures involved," and seismic events at The Geysers correlate with injection activity [34]. Induced seismicity led Basel to suspend and then cancel its project [34][18], and the 2017 Pohang earthquake may have been linked to the Pohang EGS project, after which all research activities stopped in 2018 [22]; a dedicated study assessed whether the Mw 5.4 Pohang event was induced [6]. The counter-argument in the corpus is the Australian government's position that hydrofracturing-induced seismicity risks are low relative to natural earthquakes, can be reduced by careful management and monitoring, and "should not be regarded as an impediment to further development," with the caveat that seismicity varies site to site and should be assessed before large-scale injection [34]. Mitigation practice exists — e.g., a designed and implemented traffic-light system for deep geothermal well stimulation in Finland [6]. 2. *Reservoir creation and performance remain hard to predict.* A large modelling literature exists precisely because heat-extraction performance depends on

fracture network geometry, natural fractures, thermo-hydro-mechanical(-chemical) coupling, well layout and boundary conditions [12][15][16][25][10]. NLR frames subsurface characterisation as necessary "to ensure accurate, onsite understanding of EGS to allow for optimization of fracture networks and strategic development and operational decisions" [1]. 3. *A long history of stalled or curtailed projects.* Fenton Hill ended with budget cuts [14]; Basel was cancelled [34]; Pohang was halted [22]; and a dedicated study catalogues "What Are the Challenges in Developing Enhanced Geothermal Systems (EGS)? Observations from 64 EGS Sites" [27] — a corpus signal that difficulty is widespread, although the specific findings are not reproduced here. 4. *Cost credibility.* The gap between the 2006 MIT projection and the 2022 Earthshot baseline implies EGS costs have historically been optimistically projected [17][23].

Closed-loop / advanced geothermal.

1. *Heat-transfer area is the binding constraint.* The corpus states the challenges plainly: "potentially complex downhole completions and creating sufficient area for heat transfer with the surrounding rock" [29]. Because CLG does not stimulate a fracture network [29][1], it forgoes the very mechanism EGS uses to enlarge rock-fluid contact. 2. *Unproven at scale.* The corpus describes CLG designs, attractions and NLR optimisation research [29][19][21], but provides no field demonstration of electricity output — in contrast to EGS's Soultz grid connection [18], Cooper Basin 25 MW [8], and Fervo commercial-scale demonstration [20]. 3. *Advantages are conditional/theoretical.* "Can theoretically be applied anywhere" and "opportunities for optimization and scale-up with technological improvements" are forward-looking claims, not demonstrated results [29].

Conventional hydrothermal.

1. *Fundamental siting limitation.* Traditional geothermal power operates "only where naturally occurring heat, water, and rock permeability are sufficient" [4], and most accessible geothermal energy lies in dry, impermeable rock [2 via 4]. 2. *Even hydrothermal fields need stimulation.* Much of the project table consists of permeability-recovery and stimulation interventions at operating hydrothermal fields — chemical stimulation at Berlín, Los Azufres, Leyte, Tiwi, Mindanao, Bacman; permeability recovery at Soda Lake; acid stimulation at Salak; scale removal for calcium carbonate [2][5][11][13] — indicating that natural permeability degrades or is insufficient in practice. This blurs the EGS/hydrothermal boundary rather than favouring either. 3. *Hydrothermal is also seismically implicated.* Seismic events at The Geysers geothermal field correlate with injection activity [34].

A note on superhot rock (SHR) as a fourth option. Conventional plant technologies (binary, flash, dry steam) typically need 100–350 °C, found at depths below 3 km; where fluids reach supercritical conditions (>400 °C), estimated energy output is up to 10× that of a normal geothermal well — but SHR "systems have not yet been harnessed for power production due to significant technical and technological challenges," including drilling methods, materials and reservoir monitoring [30]. SHR is therefore the highest-upside and least-proven of the options in this corpus [30].

Recoverable resource base

United States (the only quantified geography in the corpus). The 2006 MIT report, funded by DOE and described as the most comprehensive EGS analysis to date, found [17]:

- Total US EGS resource at 3–10 km depth: **over 13,000 zettajoules (ZJ)**.
- **Extractable**: over **200 ZJ**, rising to **over 2,000 ZJ with better technology**.
- All geothermal resources including hydrothermal and geo-pressured: **14,000 ZJ**, roughly **140,000× US primary energy use in 2005**.
- **Recoverable with then-current technology**: **1.2–12.2 TW** across conservative and moderate scenarios.
- **Deployable capacity**: **100 GWe or more by 2050** with a $1 billion R&D investment over 15 years.

Note the internal structure of these numbers: the "13,000 ZJ in place → >200 ZJ extractable" step is a ~1.5% recovery ratio, and the "recoverable" range spans a factor of ten (1.2–12.2 TW) purely on scenario assumptions [17]. Both features mean headline in-place figures are poor guides to deliverable capacity.

Globally. The corpus does not contain a global EGS resource estimate. It offers only qualitative framing — EGS "provide tremendous opportunity for harvesting the vast, untapped heat stored in the shallow continental crust around the world" [1], and EGS is "apparently feasible anywhere in the world, depending on the resource depth" [8] — plus one regional datum: Cooper Basin, Australia has the potential to generate 5,000–10,000 MW [8]. EGS systems are under development in Australia, France, Germany, Japan, Switzerland and the United States [8], and the global project table spans Iceland, Germany, UK, France, USA, Sweden, Japan, Austria, Russia, Philippines, Australia, El Salvador, Indonesia, Mexico, Switzerland, Italy, South Korea, China and Hungary [2][6][22]. **A global technically/economically recoverable EGS resource number is not present in the evidence provided.**

Sensitivity to depth, temperature and cost thresholds

Depth. The MIT resource base is defined over a specific depth window, 3–10 km, so the resource total is definitionally depth-bounded [17]. Depth also sets what is physically reachable: advanced drilling can penetrate hard crystalline rock to depths up to or exceeding 15 km, accessing rock at 400 °C and above because temperature increases with depth [8]. EGS feasibility is explicitly conditioned on depth ("feasible anywhere in the world, depending on the resource depth") [8]. Depth interacts with cost because drilling cost is one of the four dominant cost sensitivities [17], and geothermal drilling cost curves are an active area of NLR analysis [24].

Temperature. Resource temperature is the first of MIT's four cost-sensitivity factors [17]. Geology that preserves temperature raises value: good EGS locations are typically over deep granite covered by a 3–5 km layer of insulating sediments that slow heat loss [8]. At the top end, the temperature leverage is very large — supercritical fluids above 400 °C could yield up to 10× the energy of a normal geothermal well, which "could greatly improve the economics of geothermal energy development" [30] — while conventional conversion technologies are matched to 100–350 °C [30].

Cost thresholds. MIT tied resource realisation directly to cost drivers — temperature, fluid flow through the system, drilling costs, and power conversion efficiency [17] — and to an R&D spend threshold ($1B over 15 years for ≥100 GWe by 2050) [17]. The DOE Earthshot then set an explicit price threshold of $45/MWh by 2035, a 90% reduction [23]. Notably, the corpus quantifies *technology-improvement* sensitivity for resource size — extractable resource rising from >200 ZJ to >2,000 ZJ, a tenfold increase, "with better technology" [17] — but does **not** provide the supply-curve elasticities (MW or ZJ recoverable as a function of specific $/MWh, depth in km, or °C cut-offs) that a rigorous sensitivity analysis would require.

Evidence gaps to flag

The corpus does not contain: (i) any numerical LCOE, capex or capacity-factor values for EGS, CLG or hydrothermal — only the statement that such values exist in SAM, GeoPHIRES and the 2024 Electricity ATB [24][28]; (ii) any global recoverable EGS resource estimate [see above]; (iii) any quantified CLG resource base, cost, or demonstrated output [29]; (iv) any explicit head-to-head EGS/AGS/hydrothermal cost comparison with reported results — the one such claim rests on a bibliographic title only [15]; (v) quantified water-use, land-use or emissions comparisons, beyond the qualitative statement that CLG "lowers water consumption" relative

What policy, tax and regulatory incentives currently support EGS in

The corpus documents a reasonably clear picture of **U.S. federal appropriations, tax credits and cost-target programs** for EGS, plus scattered evidence of **foreign public funding and licensing**. It is, however, largely silent on the permitting, land-access and environmental-review questions posed in the second half of the prompt. Where the corpus does bear on regulation, the operative constraint it documents is not NEPA or leasing but **induced seismicity and the project-level social/regulatory license it governs**. I flag the gaps explicitly rather than filling them.

U.S. federal tax incentives and how the IRA changed them

The single tax-policy datum in the corpus is that the **Inflation Reduction Act extended the production tax credit (PTC) for renewable energy sources, including geothermal, until 2024, and included geothermal energy in the new technology-neutral Clean Electricity PTC beginning in 2024** [32]. That is meaningful for EGS economics in two ways that the corpus supports indirectly: EGS is a **baseload resource producing power at a constant rate** [8] with an expected **economic lifetime of 20–30 years** [8], so a per-MWh production credit accrues at a high and predictable capacity factor over a long operating life — a better fit for a generation credit than for intermittent resources. Beyond that, the corpus contains **no evidence** on: the parallel investment tax credit (48/48E) election, credit transferability or direct pay, bonus adders (energy communities, domestic content), the intangible drilling cost treatment sometimes claimed by geothermal developers, or **any tax-law change enacted after the IRA**. The prompt's reference to "subsequent tax law changes" cannot be answered from this corpus; asserting anything about post-IRA statutory amendments would be invention.

U.S. federal appropriations, demonstration programs and cost targets

Federal support has moved through three distinguishable waves in the record:

1. **Late-2000s R&D and stimulus.** In 2009 the U.S. Department of Energy issued two Funding Opportunity Announcements related to enhanced geothermal systems, together offering **up to $84 million over six years** [33]; DOE's own announcement was framed as an intent to "Invest up to $84 Million in Enhanced Geothermal Systems" [35]. DOE then opened a further FOA in 2009 using **American Reinvestment and Recovery Act stimulus funding of $350 million, of which $80 million was aimed specifically at EGS projects** [33], within a broader announcement of "Over $467 Million in Recovery Act Funding for Geothermal and Solar Energy Projects" [35].
2. **A dedicated federal field laboratory.** DOE's Geothermal Technologies Office issued a **Notice of Intent for an EGS Observatory in February 2014** [35], which matured into the **Utah Frontier Observatory for Research in Geothermal Energy (FORGE), described as an international laboratory for EGS technology development** [6][20]. FORGE functions as publicly funded de-risking infrastructure: national-laboratory researchers are partnering with Utah FORGE to test **chemical stimulation concepts intended for use by the global EGS community**, in place of or alongside mechanical stimulation [1].
3. **Infrastructure-era demonstrations and an explicit cost target.** The **Infrastructure Investment and Jobs Act authorized $84 million to support EGS development through four demonstration projects** [32]. In **September 2022** the Geothermal Technologies Office announced an **"Enhanced Geothermal Shot"** under the Energy Earthshots campaign, with the goal of **cutting EGS costs by 90%, to $45/MWh by 2035** [23][32-adjacent context in 23]. A national-laboratory technical report on the **Enhanced Geothermal Shot (2023)** sits in the same publication stream [24].

Federal RD&D has also been directed at the cost drivers most sensitive for EGS. The MIT 2006 assessment identified EGS costs as sensitive to four factors — **resource temperature, fluid flow through the system, drilling costs, and power-conversion efficiency** [17] — and current public research is aimed squarely at two of them: **drilling analysis and techno-economic modeling to optimize deep drilling efficiencies** [1], with published **geothermal drilling cost-curve updates** [24]. Public co-funding has also flowed to non-electric EGS applications: national-lab researchers are partnering with **Cornell University on EGS reservoir models and techno-economic simulations for campus heating** [1], and the corpus records a **$7.2 million grant for exploratory research into Cornell's Earth Source Heat** project [35].

How these incentives are translated into project economics

The corpus shows the mechanism by which policy is converted into bankability analysis rather than quantifying it. Publicly developed, open-source techno-economic tools — **the System Advisor Model (SAM) and GEOPHIRES** — are used to estimate site-based capital costs, O&M and levelized cost of electricity for hydrothermal or EGS projects, with SAM's geothermal module explicitly incorporating **financial models, hourly performance simulations, and policy impacts** [28]. GEOPHIRES adds dynamic modeling of reservoir performance and heat extraction over time for both electricity and direct-use applications [28]. The **Electricity Annual Technology Baseline (2024)** publishes projected LCOEs for EGS together with the key cost and performance assumptions — **capex, capacity factor, well productivity, decline and O&M** [24]. These outputs are described as informing **geospatial analysis, electricity grid capacity-expansion models, and next-generation technology cost targets**, i.e. as groundwork for **de-risking investment** and supporting commercialization [28].

Two benchmark numbers frame the gap that incentives are meant to close: MIT's 2006 claim that EGS **could** produce electricity for as low as **3.9 cents/kWh** [17] (with 100 GWe or more potentially available in the U.S. by 2050 on a $1 billion/15-year R&D investment [15][17]), versus DOE's 2022 premise that a **90% cost reduction is still required** to reach $45/MWh by 2035 [23]. The corpus does not reconcile these; on its face the Earthshot baseline implies costs far above the 2006 aspiration, and readers should treat the 2006 figure as a modeled potential rather than a realized price.

A technology-cost lever the corpus does document is the transfer of oil-and-gas capability: **long horizontal laterals and multi-stage hydraulic fracturing applied to geothermal settings** [1][14 context], with Fervo Energy reported as having **hit a milestone in using oil drilling technology to tap geothermal energy** [20] and a **commercial-scale demonstration of a first-of-a-kind EGS** documented in 2023 [20]. Utah is described in press coverage as a possible home to the **world's largest enhanced geothermal plant** [20] [35]. These are technology, not policy, developments, but they are the supply-side complement to the tax and demonstration incentives above.

International policy, funding and licensing support

- **United Kingdom.** The United Downs Deep Geothermal Power Project in Cornwall is documented as a supported deep-geothermal development [20][35], with **Geothermal Engineering Ltd receiving £15 million in funding for deep geothermal in the UK** (2023) [35] and a project update presented to the European Geothermal Congress in 2022 [35]. The corpus also carries a BBC item describing **Earth's heat producing electricity for homes in a UK clean-energy first** [35]. The corpus does not identify the specific UK support mechanism (e.g., Contracts for Difference), so the funding channel cannot be characterized further.
- **European Union.** The EU's EGS R&D project at **Soultz-sous-Forêts, France, connects a 1.5 MW demonstration plant to the grid** and explored multiple stimulated zones and triplet well configurations [18] — an example of EU-level R&D funding producing a grid-connected demonstration. The **DESTRESS H2020** programme also appears in the record in connection with Pohang [20][35], and Icelandic/EU work includes the **DEEPEGS** demonstration programme, including the Reykjanes IDDP-2 well and published "lessons learned" on project management [20].
- **Portugal.** In **December 2008 the Portuguese government awarded an exclusive license to Geovita Ltd to prospect and explore geothermal energy**, covering roughly 500 km² studied jointly with the University of Coimbra [18]. This is the corpus's clearest example of a **licensing/land-access regime abroad** — exclusive exploration concessions rather than competitive leasing.
- **Australia.** Australian government policy posture on stimulation risk is documented (see below) [34], and the country hosted what the corpus calls **the world's largest EGS project, a 25 MW demonstration plant in Cooper Basin** [8], alongside multiple Cooper Basin, Olympic Dam and Paralana stimulation campaigns [2].
- **South Korea.** The **Pohang EGS project began in December 2010 with a 1 MW goal** [22], and appears in the DESTRESS record [20][35].

The binding regulatory constraint in the evidence: induced seismicity

The corpus's regulatory story is dominated by seismicity, not by land or environmental review. **Induced seismicity is common in EGS because of the high pressures involved** [34], and seismic events at the

Geysers field in California correlate with injection activity [34]. Two projects show how this translates into hard regulatory outcomes:

- **Basel, Switzerland:** induced seismicity **led the city to suspend and later cancel the project** [34], and separately is recorded as having **led to the cancellation of the EGS project there** [18]; the Basel 1 system itself is characterized in the peer-reviewed literature [13].
- **Pohang, South Korea:** the **2017 Pohang earthquake may have been linked to the project's activity, and all research activities were stopped in 2018** [22]; a *Science* paper assesses whether the **Mw 5.4 event was induced** [6].

Against this, the corpus records a contrasting official position: according to the **Australian government**, risks from "hydrofracturing induced seismicity are low compared to that of natural earthquakes, and can be reduced by careful management and monitoring," and "should not be regarded as an impediment to further development," while noting that induced seismicity **varies from site to site and should be assessed before large-scale fluid injection** [34]. This is a genuine divergence in regulatory posture between the Swiss/Korean project outcomes and the Australian policy assessment, and both are cited here.

The mitigation architecture that has emerged is procedural rather than statutory in the record: a **traffic-light system for deep geothermal well stimulation was designed and implemented in Finland** [6], and **long-term induced-seismicity monitoring has been maintained at the Insheim site in Germany** [13]. Reservoir-design choices also interact with this constraint: **closed-loop geothermal requires no reservoir stimulation, which limits induced-seismicity risk and lowers water consumption** [29] — a relevant comparison for regulators weighing stimulation-based EGS against alternatives, though closed-loop faces its own challenges in complex downhole completions and achieving sufficient heat-transfer area [29].

Permitting, land access and NEPA: evidence is missing

The corpus contains **no evidence** on:

- NEPA review durations, categorical exclusions for geothermal exploration or observation wells, or any programmatic environmental impact statement;
- **Bureau of Land Management geothermal leasing** procedures, lease sale cadence, royalty rates, or federal-land access reform;
- state-level permitting processes, well-permitting authorities, water-rights administration, or interconnection/transmission permitting;
- quantified permitting timelines for any EGS project, in the U.S. or abroad;
- pending legislative or administrative permitting-reform proposals.

The closest the corpus comes to a land-access mechanism is Portugal's **exclusive prospecting/exploration license** award [18], and the closest it comes to a siting constraint is the observation that EGS is **feasible anywhere in the world depending on resource depth**, with favorable sites typically over deep granite beneath 3–5 km of insulating sediment [8], plus resource-mapping work on **temperature at depth and EGS potential** and **high-resolution maps of worldwide geothermal resource potential** with a national geospatial interface for developers [31]. Site-screening tools of this kind reduce exploration risk but are not a substitute for permitting evidence.

Similarly, while the record shows induced seismicity being managed through monitoring and traffic-light protocols [6][13][34], it does not describe the statutory or regulatory instruments (e.g., seismic-risk permitting conditions, liability rules, insurance requirements) through which those protocols are imposed. Any account of "reforms underway" in U.S. permitting, categorical exclusions or BLM leasing would have to come from outside this corpus.

Bottom line

Public support for EGS in the record is a three-legged structure: a **production-based tax credit that the IRA extended and then folded into a technology-neutral Clean Electricity PTC from 2024** [32]; **appropriated demonstration and R&D money** — roughly $84 million in 2009 FOAs, $80 million of ARRA funds earmarked for EGS within a $350 million package, and a further $84 million authorized by the IIJA for four demonstration projects [33][32] — plus a dedicated field laboratory at Utah FORGE [6][20]; and an **explicit government cost target of $45/MWh by 2035, a 90% reduction** [23], operationalized through open-source techno-economic tools that embed policy effects into LCOE estimates [28][24]. Abroad, support appears as EU/national R&D grants and demonstrations (Soultz, DEEPEGS, DESTRESS) [18][20], direct national funding in the UK (£15 million to GEL) [35], and exclusive exploration licensing in Portugal [18]. The dominant regulatory risk documented is **induced seismicity**, which has already terminated projects in Basel and Pohang [34][18][22] even as Australian authorities judge the risk manageable and not an impediment [34]. On U.S. permitting, NEPA, categorical exclusions and BLM leasing, this corpus provides no basis for analysis.

What are the induced seismicity risks of EGS, what lessons come from

Induced seismicity — earth tremors caused by human activity — is an intrinsic feature rather than an incidental side-effect of enhanced geothermal systems, because EGS depends on raising fluid pressure in rock to create flow paths. The Wikipedia overview states plainly that "seismicity is common in EGS, because of the high pressures involved" [34]. The physical mechanism is described in the same source: permeability is enhanced by hydro-shearing, "pumping high-pressure water down an injection well into naturally-fractured rock," which "increases the fluid pressure in the rock, triggering shear events that expand pre-existing cracks and enhance the site's permeability" [8]. In other words, the shear events *are* the stimulation product; the engineering question is controlling their magnitude, not eliminating them. Beyond stimulation, ongoing operations can also be seismogenic: seismicity events at The Geysers geothermal field in California are correlated with injection activity [34].

The breadth of stimulation techniques used across the global EGS fleet — hydraulic, chemical, thermal, carbon-based, and explosive methods [2][7] — implies that seismic risk profiles differ by method, but the corpus does not provide a comparative quantification of seismic hazard by stimulation type; that evidence is missing.

Two framings of the residual risk appear in the corpus and should be reported side by side. The first emphasises project-ending hazard: induced seismicity in Basel "led the city to suspend its project and later cancel the project" [34], and the Wikipedia EU section separately records that "induced seismicity in Basel led to the cancellation of the EGS project there" [18]. The second is more permissive: according to the Australian government, risks associated with "hydrofracturing induced seismicity are low compared to that of natural earthquakes, and can be reduced by careful management and monitoring" and "should not be regarded as an impediment to further development" [34]. The reconciling statement in the same source is that "induced seismicity varies from site to site and should be assessed before large scale fluid injection" [34] — i.e., site-specific pre-injection assessment is the operative principle, not a blanket judgement of acceptability.

Lessons from Basel

Basel, Switzerland, began in 2006 as a hydraulically stimulated EGS project [2]. Its technical characterisation is documented in the peer-reviewed literature ("Characterisation of the Basel 1 enhanced geothermal system," *Geothermics*, 2008) [13]. The outcome recorded in the corpus is unambiguous: induced seismicity caused the city first to suspend and then to cancel the project [34][18]. The transferable lesson available from this evidence is that in a dense urban setting, felt seismicity can terminate a project on social and regulatory grounds even where the reservoir engineering is otherwise documented and analysable [13][34]. The corpus does not give the magnitude of the Basel events, damage claims, or the specifics of the regulatory decision — that detail is absent and should not be assumed.

Lessons from Pohang

The Pohang EGS project in South Korea started in December 2010 with a goal of producing 1 MW [22]. The corpus reports that "the 2017 Pohang earthquake may have been linked to the activity of the Pohang EGS project" and that "all research activities were stopped in 2018" [22]. The scientific basis for the linkage is a 2018 *Science* paper assessing "whether the 2017 Mw 5.4 Pohang earthquake in South Korea was an induced event" [6]; the Pohang case was also taken up within the EU DESTRESS programme [35]. Note the epistemic status carefully: the corpus states the link as probable/under assessment ("may have been linked") rather than as settled fact [22][6]. The lesson supported by this evidence is that EGS stimulation has been associated with an event of damaging magnitude (Mw 5.4), not merely microseismicity, and that such an event was sufficient to halt a national research programme entirely [22][6].

St. Gallen and other monitored sites

For St. Gallen, Switzerland, the corpus contains only the project documentation — "The St. Gallen Project: Development of Fault Controlled Geothermal Systems in Urban Areas," presented at the World Geothermal Congress 2015 [13][3] — cited alongside Basel and other cases as part of the record of EGS development experience. The title indicates the project targeted fault-controlled permeability in an urban area [13], which is the same combination of hazard drivers (pre-existing faults plus population exposure) present at Basel. However, **the corpus does not contain any account of a seismic event, its magnitude, or the project's outcome at St. Gallen** any specific "lesson from St. Gallen" beyond the fault-controlled, urban character of the project would be unsupported by this evidence base.

Relevant adjacent experience in the corpus includes long-term monitoring of induced seismicity at the Insheim geothermal site in Germany, published in the *Bulletin of the Seismological Society of America* (2018) [13], which demonstrates that multi-year seismic surveillance of operating EGS-type sites is an established practice. The Northwest Geysers EGS Demonstration Project similarly generated documented characterisation of reservoir response to injection [3].

Traffic-light systems and mitigation protocols

The corpus supports the existence and engineered design of traffic-light protocols but does not support a claim that any specific protocol is now globally "standard."

- A traffic light system for deep geothermal well stimulation was designed and implemented in Finland, documented in the *Journal of Seismology* (Ader et al., 2019), covering both design and implementation for a deep stimulation campaign [6]. This is the only explicit traffic-light system in the evidence base.
- The general mitigation principle stated in the corpus is that seismicity risk "can be reduced by careful management and monitoring," and that because risk varies site to site, it "should be assessed before large scale fluid injection" [34].
- Long-term seismic monitoring of operating sites is demonstrated at Insheim [13], and reservoir/injection-response characterisation is demonstrated at Northwest Geysers [3].
- An alternative mitigation route is technological rather than procedural: closed-loop geothermal (CLG) is attractive in part because "reservoir stimulation is not required, which limits the risk of induced seismicity and lowers water consumption" [29]. This is a design-level avoidance of the hazard rather than a management protocol, and it comes with its own trade-offs — "potentially complex downhole completions and creating sufficient area for heat transfer with the surrounding rock" [29].

What the corpus does **not** contain: magnitude thresholds, amber/red trigger levels, shut-in and bleed-off procedures, standardised seismic hazard assessment frameworks, insurance or liability arrangements, community engagement requirements, or any regulatory instrument mandating traffic-light systems in a given jurisdiction.

Environmental footprint: water, land, emissions

This is the weakest-evidenced part of the question, and the honest answer is that the corpus is largely silent. What can be said:

Water. EGS is a circulating-fluid technology: “water passes through the fractures, absorbing heat until forced to the surface as hot water,” is converted to electricity via a steam turbine or a binary plant that cools the water, and “the water is cycled back into the ground to repeat the process” [8]. This recirculating architecture is relevant to net consumption but is not quantified anywhere in the corpus. One comparative signal exists: CLG “lowers water consumption” relative to systems requiring reservoir stimulation [29], which implies EGS water consumption is materially higher than closed-loop alternatives, though no volumes (e.g., L/MWh, or make-up water for stimulation and circulation losses) are given. A relevant technical note is that hydro-shearing does not require hydraulic fracturing proppants to keep fractures open, provided injection pressure is maintained [8] — reducing the chemical/materials burden relative to proppant-based fracturing, though the corpus also records that chemical stimulation is used at many sites, including planned chemical stimulation testing at Utah FORGE [1][2].

Land use. The corpus contains no land-use intensity data for EGS (no acres/MW, well-pad footprints, or comparisons). Indirectly, the trend toward “long horizontal laterals and multi-stage hydraulic fracturing” [1] and multi-well/horizontal-well configurations [10][12] suggests surface footprint concentration at fewer pads, but this is an inference the corpus does not make. **Evidence missing.**

Emissions. The corpus contains no lifecycle greenhouse-gas figures, no non-condensable gas or H_2S data, and no air-quality data for EGS. EGS appears in the corpus within a Wikipedia “Renewable energy” series context [4] and is discussed alongside carbon storage in one feasibility study title (“Integrated technological and economic feasibility comparisons of enhanced geothermal systems associated with carbon storage”) [15], but no emissions quantity is stated. **Evidence missing.**

Comparison with other firm generation sources. The only firmness-relevant characterisation in the corpus is that “EGS plants are baseload resources that produce power at a constant rate” [8], with an expected economic lifetime of 20–30 years [8], and cost figures of as low as 3.9 cents/kWh (2006 MIT estimate) [17][10] against a DOE Earthshot target of $45/MWh by 2035 [23]. Cost and performance assumptions for EGS — capex, capacity factor, well productivity, decline, and O&M — are compiled in NREL’s Electricity Annual Technology Baseline (2024) [24], and techno-economic tools such as SAM and GeoPHIRES are used to model EGS and hydrothermal LCOE [28]. **However, the corpus provides no environmental-footprint comparison between EGS and nuclear, gas with CCS, coal, hydropower, or storage-backed renewables.** Any such comparison would be fabrication on this evidence base.

Summary of the defensible position

Induced seismicity is an inherent, mechanistically necessary consequence of pressure-driven EGS stimulation [8][34], it has demonstrably ended at least one project on urban social/regulatory grounds (Basel) [34][18] and been probably associated with a Mw 5.4 damaging event that halted a national programme (Pohang) [22][6], while an official Australian assessment maintains the hazard is low relative to natural earthquakes and manageable through monitoring [34] — a genuine conflict of framing within the corpus. Documented mitigation consists of site-specific pre-injection assessment [34], long-term monitoring [13], engineered traffic-light systems as implemented in Finland [6], and, at the design level, avoidance via non-stimulated closed-loop concepts [29]. On water, land, and emissions footprints, the corpus supports only qualitative statements (recirculating water use [8]; higher water use than CLG [29]) and contains no quantitative footprint data or cross-technology environmental comparison; that evidence would need to be sourced separately.

What is the grid and market value of EGS as firm, flexible clean

Summary of what the evidence can and cannot support

The corpus assembled for this review is overwhelmingly technical and programmatic (reservoir stimulation methods, project histories, drilling, resource assessment, and cost-target policy). It contains **no** evidence on the specific market questions posed: there are no EGS power purchase agreement (PPA) prices, no named offtakers, no data-center procurement deals, no capacity-credit or effective-load-carrying-capability studies, no dispatch/load-following operating data, and no head-to-head levelized-cost or system-value comparisons of EGS against solar-plus-storage or gas-fired generation. Those items are simply **missing from the evidence base**, and I will not infer them. What follows is what can be legitimately said, plus an explicit inventory of the gaps.

The physical basis for EGS's grid value: firmness, not (documented) flexibility

The clearest grid-relevant attribute documented is firmness. EGS plants are described as "baseload resources that produce power at a constant rate," and, unlike conventional hydrothermal, EGS "is apparently feasible anywhere in the world, depending on the resource depth" [8]. Two implications follow directly from that language, and only those two:

- **Firm output profile.** Constant-rate production is the operating mode described in the evidence [8]. This is the physical property that underpins claims of high capacity value in the wider literature — but the corpus contains no capacity-value or capacity-credit quantification for EGS, so the magnitude of that value is not established here.
- **Siting flexibility rather than dispatch flexibility.** The distinguishing economic feature emphasized is geographic reach — EGS is not confined to naturally convective hydrothermal fields [4][8] — and NREL's framing is that new reservoir designs "could significantly reduce upfront project risk and allow for the development of geothermal anywhere in the United States" [19]. Siting breadth matters for grid value (resources can be located near load or in constrained zones), but the corpus does not document any locational-value or transmission-deferral analysis.

Crucially, the corpus provides **no evidence that EGS plants are operated in a load-following or flexible mode**, and no evidence on ramp rates, turndown, minimum stable output, or thermal-storage-style curtailment-and-recovery operation. The one adjacent signal is that active flow-control research exists at the reservoir level: work on "downhole flow control" as a key for developing EGS in horizontal wells [10] and on "real-time temperature monitoring and flow control for improved heat extraction" [12]. These are heat-extraction optimization concepts in the modeling literature, not demonstrated grid-following dispatch, and should not be read as evidence of market-facing flexibility.

Two further physical/economic parameters bear on value stacking:

- **Asset life.** EGS plants "are expected to have an economic lifetime of 20–30 years" [8]. This is a shorter horizon than is often assumed for firm thermal assets and directly affects levelized cost and financing.
- **Thermal drawdown risk.** The dominant theme in the modeling corpus is heat-extraction performance and decline over time — e.g., multi-well injection configurations, fracture-network geometry, and coupled thermo-hydro-mechanical effects on extraction performance [15][25][12][16]. NREL's GEOPHIRES tool is described as offering dynamic modeling "with a strong focus on reservoir performance and heat extraction over time" [28]. Sustained output — not instantaneous flexibility — is therefore the modeled risk to firm value.

Cost signals, which are the only quantitative "market value" anchors available

The corpus supplies three cost anchors, and they are of very different vintages and natures. They are cost *estimates and targets*, not contracted prices.

1. **MIT (2006):** the report "claimed that EGS could produce electricity for as low as 3.9 cents/kWh," with costs sensitive to four factors — resource temperature, fluid flow through the system, drilling costs, and power

conversion efficiency [17][11-context: see 17]. Note this is a 17-year-old modeled floor, expressed as a best case ("as low as"), not an achieved price [17].

2. **DOE Enhanced Geothermal Shot (2022):** the Geothermal Technologies Office set a goal "to reduce the cost of EGS by 90%, to $45/megawatt hour by 2035" [23]. The 90% reduction implies the program's assumed 2022 baseline was roughly an order of magnitude above $45/MWh — i.e., far above the MIT 2006 figure of ~$39/MWh [17][23]. **This is a genuine tension in the evidence:** a 2006 study asserting sub-4-cent/kWh potential sits alongside a 2022 federal program premised on EGS currently costing about ten times its 2035 target [17][23]. The most defensible reading is that the MIT number was a technology-potential estimate rather than a then-achievable cost, but the corpus does not reconcile them explicitly.
3. **NREL Annual Technology Baseline (2024):** the Electricity ATB "includes projected levelized costs of electricity for enhanced geothermal systems (EGS), along with key cost and performance assumptions such as capex, capacity factor, well productivity, decline, and O&M" [24]. This is the corpus's most current and most decision-relevant cost source — and notably its parameter list (capacity factor, well productivity, decline) again frames EGS value as a firm, declining-output resource. **The actual ATB numerical values are not reproduced in the corpus**, so no LCOE figure can be quoted here.

For competitive positioning, the corpus offers only one comparative statement, and it is weak: an integrated technological and economic feasibility comparison indicates that EGS "can be competitive with closed-loop/advanced geothermal (AGS) and conventional hydrothermal systems under specific conditions" [15][27]. That comparison is *within* the geothermal family. **There is no evidence in this corpus comparing EGS to solar-plus-storage, wind, gas combined cycle, or gas peakers** on cost, capacity value, or system value.

Policy-driven revenue, which substitutes for market evidence

Where the corpus is silent on PPAs, it does document the policy support that shapes contractable economics in the U.S.:

- The Inflation Reduction Act "extended the production tax credit (PTC) for renewable energy sources (including geothermal) until 2024 and included geothermal energy in the new Clean Electricity PTC to begin in 2024" [32]. Tax-credit eligibility is the mechanism by which EGS output gains value beyond energy-market revenue, but the corpus gives no credit values, no basis-adjustment detail, and no evidence of how developers monetize it.
- Direct public capital has been substantial and repeated: the Infrastructure Investment and Jobs Act "authorized $84 million to support EGS development through four demonstration projects" [32]; DOE issued two 2009 funding announcements offering "up to $84 million over six years" [33]; and a further 2009 Recovery Act announcement of "$350 million, including $80 million aimed specifically at EGS projects" [33][35]. Outside the U.S., a UK deep-geothermal developer received "£15 million funding for deep geothermal in UK" [35].
- The persistence of demonstration-stage public funding through 2022, alongside a 90%-cost-reduction Earthshot target [23], is itself the strongest available indicator that EGS was not yet competing unsubsidized in energy markets at that time.

Offtakers, PPAs, and data-center demand: evidence is absent

On the second half of the question, the corpus permits almost nothing. Specifically:

- **No PPA prices, terms, tenors, shapes (baseload vs. 24/7 hourly-matched), or counterparties appear anywhere in the evidence.**
- **No data-center or hyperscaler offtake agreements are documented.** No corporate offtaker is named in any capacity.
- The only documented offtake-like arrangements are non-commercial and thermal, not electric: NREL "is partnering with Cornell University to develop EGS reservoir models and run techno-economic simulations of utilizing EGS to provide heating to the university campus" [1], supported by Cornell's own campus decarbonization program and a "$7.2M grant funds exploratory research into Earth Source Heat" [35]. This

indicates a district-heating/campus-thermal demand pathway, which is a genuinely distinct market from firm power, and NREL's GEOPHIRES explicitly covers "direct-use geothermal applications" alongside electricity [28]. But it is a research partnership, not a priced offtake.
- Grid-facing commercialization is evidenced only obliquely: a commercial-scale EGS demonstration is documented as a preprint/technical milestone [20], Fervo's use of oil-and-gas drilling technology is documented as a technical "milestone" report [20][1], a Utah project is described in press as potentially "the world's largest enhanced geothermal plant" [20], a 1.5 MW Soultz demonstration plant is "connect[ed] … to the grid" [18], and a UK report describes "Earth's heat to produce electricity for homes in UK clean energy first" [35]. None of these sources, as excerpted, disclose contract prices or buyers.
- On scale and demand-side pull, the corpus offers only supply-side potential: 100 GWe or more "could be available by 2050 in the United States" with a $1 billion R&D investment over 15 years, and recoverable resources of 1.2–12.2 TW [17]; Cooper Basin alone has "the potential to generate 5,000–10,000 MW," with the world's largest EGS project there being a 25-megawatt demonstration plant [8][15-context: see 8]. The gap between a 25 MW demonstration and multi-gigawatt potential [8] is the honest characterization of current market scale. **No load-growth or data-center demand forecasts appear in the corpus.**

How the value question is currently being answered analytically (rather than by markets)

Because contracted-price evidence is absent, the corpus's most useful contribution is to show *how* grid and market value for EGS is being estimated. NREL "plays a leading role in advancing techno-economic analysis for next-generation geothermal technologies," using open-source tools including the System Advisor Model (SAM) and GEOPHIRES [28]. SAM "enables users to estimate site-based capital costs, operations and maintenance costs, and levelized costs of electricity of hydrothermal or EGS technologies," and its geothermal module supports "detailed techno-economic simulations of geothermal projects incorporating financial models, hourly performance simulations, and policy impacts" [28]. Critically for the grid-value question, these efforts "have informed advanced geospatial analysis, **electricity grid capacity expansion models**, and next-generation technology cost targets," with the stated purpose of "de-risking investment in innovative geothermal solutions and supporting their successful commercialization" [28]. A geothermal module has also been developed in reV [24], and drilling cost curves are being updated [24].

In other words: EGS's grid and market value is, on this evidence, being established through capacity-expansion modeling and cost-target setting [28][23][24] rather than revealed through observable PPA transactions. Hourly performance simulation capability exists in SAM [28], which is the necessary precondition for assessing capacity value and any flexible operation — but no results from such analyses are present in the corpus.

Risk factors that bear directly on contractability

Two documented risks are material to any offtaker's willingness to sign a firm contract:

- **Induced seismicity.** "Seismicity is common in EGS, because of the high pressures involved," seismic events at the Geysers correlate with injection activity, induced seismicity "led the city [of Basel] to suspend its project and later cancel the project," and the 2017 Pohang earthquake "may have been linked to the activity of the Pohang EGS project," after which "[a]ll research activities were stopped in 2018" [34][22]. Mitigation practice exists — traffic-light stimulation protocols have been designed and implemented in Finland [6] — and the Australian government's position is that hydrofracturing-induced seismicity risk "is low compared to that of natural earthquakes … and should not be regarded as an impediment to further development," while also noting that seismicity "varies from site to site and should be assessed before large scale fluid injection" [34]. This is a project-cancellation-scale risk, directly relevant to delivery certainty under a PPA [34][22].
- **Competition from closed-loop designs on exactly the firmness dimension.** Closed-loop geothermal is attractive partly because "[h]eat production can be estimated with relatively high confidence" and "[r]eservoir stimulation is not required, which limits the risk of induced seismicity and lowers water consumption" [29]. For a buyer purchasing firm output, higher-confidence production forecasting and lower seismicity risk

are competitive advantages against EGS [29], though CLG faces its own challenges of "potentially complex downhole completions and creating sufficient area for heat transfer" [29]. Superhot-rock resources offer up to ten times the energy output of a normal geothermal well but "have not yet been harnessed for power production due to significant technical and technological challenges" [30].

Bottom line

On the available evidence, EGS's grid value proposition rests on two documented attributes — constant-rate, baseload output [8] and near-universal siting feasibility subject to depth [8][4] — combined with a large technical resource base [17]. Its market value is currently expressed almost entirely through modeled costs and public cost targets: a 2006 modeled floor of 3.9 cents/kWh [17], a 2035 federal target of $45/MWh representing a 90% reduction from a 2022 baseline [23], and ATB projected LCOEs whose values are not reproduced here [24]. Policy revenue via the Clean Electricity PTC and repeated demonstration funding are documented [32][33][35].

Everything else the question asks for — quantified capacity value, demonstrated load-following or flexible dispatch, explicit competition against solar-plus-storage and gas, named offtakers, PPA prices and structures, and data-center-driven demand growth — **is not supported by this corpus.** Answering those elements would require sources outside the evidence provided.

How is EGS being financed — equity rounds, debt, tax equity, cost of

The evidence corpus contains **no direct evidence on private EGS equity rounds, project-level debt, tax-equity structures, weighted-average cost of capital, or drilling-risk mitigation instruments** (e.g., completion insurance, resource-risk guarantee funds, or exploration-risk facilities). Any characterization of those channels would be invention. What the corpus does document is (i) a public-funding and tax-credit backbone in the United States, (ii) one disclosed private/grant funding event in the United Kingdom, (iii) cost targets and cost-sensitivity findings that define the investability problem, and (iv) the analytical tools being used to de-risk investment decisions. Each is treated below, with the gaps flagged explicitly.

Public grant funding and demonstration capital

The clearest financing pathway in the evidence is federal appropriation for demonstration projects. The U.S. Department of Energy issued two Funding Opportunity Announcements in 2009 specific to enhanced geothermal systems, together offering up to $84 million over six years [33]. In the same year DOE opened a further FOA using American Reinvestment and Recovery Act stimulus funding worth $350 million in total, of which $80 million was aimed specifically at EGS projects [33]. Contemporaneous announcements referenced in the corpus confirm this posture — "DOE to Invest up to $84 Million in Enhanced Geothermal Systems" and "President Obama Announces Over $467 Million in Recovery Act Funding for Geothermal and Solar Energy Projects" [35].

More recently, the Infrastructure Investment and Jobs Act authorized $84 million to support EGS development through four demonstration projects [32]. This is grant capital directed at demonstration risk rather than at balance-sheet or project finance, and the corpus provides no evidence on how those demonstrations are co-financed by private sponsors.

Tax credits

The Inflation Reduction Act extended the production tax credit (PTC) for renewable energy sources, including geothermal, until 2024, and included geothermal energy in the new Clean Electricity PTC beginning in 2024 [32]. This is the only tax-policy instrument documented in the corpus. Critically, **the corpus contains no**

evidence on how those credits are monetized — no information on tax-equity partnership structures, transferability, credit pricing, or the share of EGS project capital stacks that tax equity supplies. The existence of a PTC is documented [32]; its financing mechanics are not.

Private and non-U.S. funding

One discrete non-U.S. funding event appears: Geothermal Engineering Limited (GEL) received £15 million in funding for deep geothermal work in the UK, in the context of the United Downs Geothermal Power Project in Cornwall, which the corpus describes as having drawn on public and private investment mechanisms including government grants and private equity participation [35]. The corpus also records that United Downs has been the subject of repeated technical reporting at the European Geothermal Congress [35] and that the UK expects Earth's heat to produce electricity for homes as a clean-energy first [35]. Beyond this single instance, the corpus provides no equity round sizes, investor identities, valuations, or debt terms for any EGS developer — including Fervo Energy, which appears in the corpus only in relation to a technical drilling milestone, not a financing event [20].

Cost of capital and the investability barriers that the evidence does support

Although cost of capital is not directly evidenced, the corpus identifies the underlying variables that drive it.

1. **Levelized cost is still far above target.**

DOE's Geothermal Technologies Office announced an "Enhanced Geothermal Shot" in September 2022 under the Energy Earthshots campaign, with the goal of reducing the cost of EGS by 90%, to $45/MWh by 2035 [23], [32-adjacent policy context]. A 90% required reduction is itself the clearest statement in the corpus that current EGS economics are not yet investable at scale on unsubsidized terms [23]. This sits in tension with the much older 2006 MIT assessment, funded by DOE, which claimed EGS could produce electricity for as low as 3.9 cents/kWh (≈$39/MWh) [17]. The two figures are not reconcilable from the corpus alone: the MIT figure was a forward-looking potential claim [17], while the 2022 Earthshot target implies realized costs roughly an order of magnitude above $45/MWh [23]. Readers should treat the 3.9 ¢/kWh number as an aspiration in a 2006 study, not an observed cost.

2. **The cost drivers are concentrated in exactly the areas lenders discount.**

MIT found EGS costs to be sensitive to four main factors: temperature of the resource, fluid flow through the system, drilling costs, and power conversion efficiency [17]. Three of these four are subsurface uncertainties resolved only after capital is sunk, which is the structural reason drilling-risk instruments matter — though, again, no such instrument is documented in this corpus.

3. **Resource and flow performance uncertainty is explicitly a modeled risk parameter.**

NREL/NLR's Electricity Annual Technology Baseline (2024) includes projected levelized costs of electricity for EGS along with key cost and performance assumptions such as capex, capacity factor, well productivity, decline, and O&M [24]. That well productivity and decline are treated as headline assumptions indicates they are material valuation levers rather than settled parameters [24].

4. **Stranding and termination risk from induced seismicity.**

Seismicity is common in EGS because of the high pressures involved [34]. Induced seismicity in Basel led the city to suspend and later cancel the project [34], [18]. The Pohang EGS project, begun in December 2010 with a 1 MW goal [22], was linked to the 2017 Pohang earthquake, and all research activities were stopped in 2018 [22]. Against this, the Australian government's position is that risks from hydrofracturing-induced seismicity are low compared with natural earthquakes, can be reduced by careful management and monitoring, and should not be regarded as an impediment to further development [34]; induced seismicity varies site to site and should be assessed before large-scale fluid injection [34]. Mitigation practice exists — a traffic-light system for deep geothermal well stimulation has been designed and implemented in Finland [6] — but the corpus shows two outright project cancellations [34], [22], which is a documented tail risk for any lender or equity underwriter.

5. Comparative competitiveness is conditional.

One integrated technological and economic feasibility comparison finds that EGS can be competitive with closed-loop/advanced geothermal (AGS) and conventional hydrothermal systems only under specific conditions [15], [27]. Closed-loop geothermal is described as attractive partly because heat production can be estimated with relatively high confidence and reservoir stimulation is not required, which limits induced-seismicity risk and lowers water consumption [29] — attributes that map directly onto financeability and that EGS does not share.

De-risking through analysis rather than through financial instruments

The corpus shows the de-risking effort concentrated on techno-economic modeling rather than on insurance or guarantee products. NREL/NLR has developed a suite of open-source modeling tools including the System Advisor Model (SAM) and GeoPHIRES [28]. SAM lets users estimate site-based capital costs, O&M costs, and levelized costs of electricity for hydrothermal or EGS technologies, supporting early-stage design and scenario testing, and its geothermal module supports detailed techno-economic simulations incorporating financial models, hourly performance simulations, and policy impacts [28]. GeoPHIRES offers dynamic modeling for hydrothermal and EGS electricity and direct-use applications with a focus on reservoir performance and heat extraction over time [28]. The stated purpose of this work is explicitly financial: these capabilities “serve as critical groundwork for de-risking investment in innovative geothermal solutions and supporting their successful commercialization,” and have informed geospatial analysis, grid capacity expansion models, and next-generation technology cost targets [28]. Related work includes Geothermal Drilling Cost Curve Updates [24] and techno-economic simulations of EGS campus heating with Cornell University [1]; Cornell’s own program has been supported by a $7.2M grant for exploratory research into Earth Source Heat [35]. Subsurface characterization and EGS-potential resource mapping are also framed as reducing pre-investment uncertainty [31], [26], [19], [21].

Evidence gap statement for financing: the corpus contains nothing on venture/growth equity round sizes or investors, nothing on project debt terms, tenor, or DSCR expectations, nothing on tax-equity or transferability pricing, no quantified discount rates or WACC for EGS, and no drilling-risk insurance, cost-overrun, or geological-risk guarantee facilities. These should be researched from sources outside this corpus.

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Supply chain, equipment and workforce constraints on scaling to tens of gigawatts

The scale gap that must be crossed

The corpus establishes the magnitude of the required build-out and the current demonstration base, and the gap between them is large. The 2006 MIT assessment estimated that with $1 billion of R&D over 15 years, 100 GWe or more could be available by 2050 in the United States, with “recoverable” resources accessible with then-current technology of 1.2–12.2 TW across conservative and moderate scenarios [17], [15-adjacent]. Against that, the corpus identifies the world’s largest EGS project as a 25-megawatt demonstration plant in Cooper Basin, Australia, with Cooper Basin’s theoretical potential put at 5,000–10,000 MW [8]. The EU’s Soultz-sous-Forêts R&D project connects a 1.5 MW demonstration plant to the grid [18], and the Pohang project targeted 1 MW [22]. Fenton Hill, the first deep full-scale EGS reservoir attempt, achieved a thermal power level of about 10 MW in a 30-day flow test before budget cuts ended the study [14]. Scaling from single- and tens-of-megawatt demonstrations [8], [18], [22], [14] to tens of gigawatts [17] is therefore a several-thousand-fold expansion in deployed capacity, and it is the physical inputs to that expansion — rigs, tools, turbines, crews — that would bind.

Drilling: the dominant binding input

Drilling is identified in the corpus both as a cost driver and as the critical technical step. MIT lists drilling costs among the four main sensitivities of EGS cost [17]. NREL/NLR states that drilling analysis and techno-economic modeling to optimize deep drilling efficiencies is “a critical step in developing EGS” [1], and maintains work on Geothermal Drilling Cost Curve Updates [24].

The depth and rock conditions define the rig and bit requirement. Advanced drilling techniques can penetrate hard crystalline rock to depths of up to or exceeding 15 km, providing access to higher-temperature rock at 400 °C and above [8]. Good EGS locations are typically over deep granite covered by a 3–5 km layer of insulating sediments [8]. Drilling long horizontal laterals in such settings has been enabled by importing oil-and-gas techniques — long horizontal laterals and multi-stage hydraulic fracturing applied to geothermal settings [1] — and Fervo Energy’s milestone is specifically framed as the use of oil drilling technology to tap geothermal energy [20]. The corpus thus supports the inference that EGS scale-up would compete with, and depend on the availability of, oil-and-gas-derived rigs, directional-drilling services, and multi-stage completion hardware [1], [20]. **The corpus does not contain rig counts, rig-day rates, utilization data, or any quantitative assessment of rig availability**, so the tightness of that constraint cannot be quantified from this evidence.

High-temperature tools, materials and completions

The corpus documents a genuine tooling frontier at high temperature. Superhot rock (SHR) systems, where fluids reach supercritical temperatures and pressures above 400 °C and estimated energy output is up to 10 times that of a normal geothermal well, “have not yet been harnessed for power production due to significant technical and technological challenges” [30]. NLR explicitly aims to address technological challenges related to SHR development including advanced drilling methods, materials, and reservoir monitoring [30] — a direct statement that high-temperature drilling and downhole hardware are not yet mature [30]. Deep, high-temperature demonstration wells such as the Reykjanes DEEPEGS well IDDP-2 [20] and the Utah FORGE international laboratory for EGS technology development [6], [20] exist precisely as testbeds for this equipment class.

Downhole flow control and zonal isolation add further tool demand: multi-stage horizontal wells with downhole flow control and multifracture stimulation have been shown to improve heat production performance [10], and real-time temperature monitoring and flow control have been studied for improved heat extraction [12]. Each of these implies sliding sleeves, packers, gauges and fiber systems rated for reservoir temperatures — for reference, Fenton Hill operated at 185 °C rock temperature at 2.6 km [14], while the SHR target is above 400 °C [30]. On the closed-loop side, acknowledged challenges include “potentially complex downhole completions and creating sufficient area for heat transfer with the surrounding rock” [29], which is a comparable completions-supply issue in an adjacent technology.

One corpus entry asserts that commercial-scale EGS deployment “highlights the need for specialized skills and advanced tools, such as high-temperature equipment, to operate effectively in deep geothermal environments” [20]. This claim should be treated cautiously: the underlying source text is a Wikipedia reference list containing the Norbeck and Latimer commercial-scale demonstration citation [20], and it does not itself present workforce or tooling data. The substantive, well-supported statement on high-temperature tooling immaturity is the SHR passage [30].

Power conversion — turbines and ORC/binary units

The corpus is thin here but not silent. Conventional geothermal power conversion — binary, flash and dry steam systems — typically requires temperatures between 100 °C and 350 °C, encountered at depths of less than 3 km [30]. EGS output heat is converted to electricity using either a steam turbine or a binary power plant system, which cools the water before reinjection [8]. Power conversion efficiency is one of MIT’s four principal cost sensitivities [17], and capacity factor is a headline assumption in the ATB EGS cost projections [24]. **However, the corpus contains no evidence on ORC/binary unit manufacturing capacity, turbine lead times, order books, heat-exchanger or working-fluid supply, or vendor concentration.** The turbine/ORC supply-chain question cannot be answered from this evidence and requires additional sourcing.

Workforce and skilled crews

The corpus provides no workforce data — no headcounts, no training-pipeline analysis, no data on directional drillers, completions engineers, or geoscientists available for geothermal work. What can be said is indirect: EGS scale-up is predicated on transferring oil-and-gas-developed techniques and technology into geothermal settings [1], [20], which implies reliance on the oil-and-gas labor pool, and the modeling and characterization workstreams described by NLR require specialized reservoir-modeling and high-performance-computing capability [19], [28], [21]. Assertions about crew availability binding a tens-of-gigawatt build should therefore be flagged as unevidenced in this corpus.

Subsurface characterization and site-by-site assessment as a throughput constraint

A less obvious scaling constraint is documented: because induced seismicity varies from site to site and should be assessed before large-scale fluid injection [34], each project carries a bespoke pre-injection assessment burden, plus monitoring regimes such as traffic-light systems [6] and long-term seismic monitoring of the kind applied at Insheim, Germany [13]. NREL/NLR frames subsurface characterization as necessary to ensure accurate, onsite understanding of EGS so that fracture networks can be optimized and development and operational decisions made strategically [1], and it produces high-resolution worldwide resource-potential maps and EGS-potential characterizations [31], [26], [19]. A survey of challenges observed across 64 EGS sites exists in the literature [27], underscoring that site heterogeneity — not a single repeatable design — has characterized the field to date [27], [2].

Summary judgment

On financing, the corpus supports only a partial picture: EGS to date has been funded principally by public grants and demonstration appropriations ($84M across two 2009 DOE FOAs [33]; $80M within a $350M ARRA tranche [33]; $84M for four IIJA demonstrations [32]; a $7.2M Cornell exploratory grant [35]; £15M to GEL in the UK [35]) plus federal tax credits extended and expanded under the IRA [32], with de-risking pursued through open-source techno-economic modeling rather than through financial risk-transfer instruments [28]. The core investability barriers evidenced are the 90% cost reduction still required to reach $45/MWh [23], concentration of cost sensitivity in subsurface variables resolved only post-capex [17], [24], and demonstrated project-termination risk from induced seismicity [34], [22], partially offset by mitigation practice [6] and by at least one regulator's view that the risk is manageable [34]. On supply chain, the best-evidenced binding constraints are deep hard-rock drilling capability and cost [8], [1], [17], [24] and high-temperature tools, materials and monitoring for the hottest resources, which are explicitly not yet mature [30]; the turbine/ORC and workforce constraints are plausible but essentially undocumented in this corpus and should be sourced separately.

What deployment scenarios and roadmaps exist for reaching

The evidence corpus contains **two explicit, quantified roadmaps** for scaling enhanced geothermal systems (EGS) — one capacity-based and one cost-based — plus a set of project-level and policy-level markers that function as de facto milestones. It does **not** contain any document that specifies a learning rate (a progress ratio or percentage cost reduction per capacity doubling), nor any year-by-year gigawatt build-out schedule. That gap should be read as a real limitation of what can be said here rather than as an absence of such analyses in the wider literature.

Roadmap 1: the MIT/DOE capacity scenario (100 GWe by 2050)

The most detailed capacity roadmap in the corpus is the 2006 MIT report funded by the U.S. Department of Energy, described as "the most comprehensive analysis to date on EGS" [17][15]. Its structure is a resource-to-deployment cascade:

- **Resource base:** U.S. EGS resources at 3–10 km depth exceed 13,000 zettajoules, of which over 200 ZJ were assessed as extractable, rising to over 2,000 ZJ "with better technology" [17]. Total geothermal resources including hydrothermal and geo-pressured were put at 14,000 ZJ, roughly 140,000 times U.S. primary energy use in 2005 [17].
- **Recoverable capacity:** between **1.2 and 12.2 TW** for conservative and moderate scenarios respectively, using then-current technology [17][6-sq06 as cited: 17].
- **Deployment scenario:** with an R&D investment of **$1 billion over 15 years**, "100 GWe (gigawatts of electricity) or more could be available by 2050 in the United States" [17][15].
- **Cost outcome:** as low as **3.9 cents/kWh** [17].

This is the corpus's only stated multi-gigawatt-to-hundred-gigawatt trajectory. Note its logic: it is an *R&D-input-to-capacity-output* scenario, not a diffusion or learning-curve model. The mechanism linking the $1B to the 100 GWe is not specified in the available text.

Roadmap 2: the cost-target roadmap (Enhanced Geothermal Shot)

In September 2022 the DOE Geothermal Technologies Office announced an "Enhanced Geothermal Shot" under the Energy Earthshots campaign, with the goal of **reducing EGS cost by 90%, to $45/MWh by 2035** [23]. This is a cost roadmap with no attached capacity target in the corpus. It implies an extremely steep cost trajectory — a factor-of-ten reduction over roughly thirteen years — but the corpus does not disclose the assumed decomposition of that reduction between drilling, stimulation, well productivity, plant capex, or financing.

Roadmap 3: project- and basin-scale scenarios

The only site-level multi-gigawatt claim is Australia's Cooper Basin, where "the world's largest EGS project is a 25-megawatt demonstration plant" and the basin is credited with "the potential to generate 5,000–10,000 MW" [8][15]. The three-orders-of-magnitude gap between the demonstration scale and the asserted basin potential is itself the central scaling question, and the corpus contains a retrospective assessment titled "What did we learn about EGS in the Cooper Basin?" [11][5], indicating the project did not proceed to that scale — though the corpus does not state the outcome explicitly.

Other sub-gigawatt anchors: the EU's Soultz-sous-Forêts project connected a **1.5 MW** demonstration plant to the grid and explored connection of multiple stimulated zones and triplet (1 injector/2 producer) well configurations [18]; Pohang, South Korea targeted **1 MW** from December 2010 [22]; Fenton Hill's second reservoir reached about **10 MW thermal** in a 30-day flow test [14].

Learning rates: the evidence is missing

No document in this corpus states a learning rate, progress ratio, or experience-curve exponent for EGS. What the corpus provides instead are the *parameters through which learning would have to act*, and the *tools used to model them*:

- MIT identified four cost sensitivities: **temperature of the resource, fluid flow through the system, drilling costs, and power conversion efficiency** [17]. Any credible learning-rate assumption must be expressed through these four levers.
- The Electricity Annual Technology Baseline (2024) contains projected levelized costs of electricity for EGS "along with key cost and performance assumptions such as capex, capacity factor, well productivity, decline, and O&M" [24]. These are the observable quantities against which a learning trajectory could be tested, but the projected values themselves are not reproduced in the corpus.

- Dedicated drilling cost curve work exists ("Geothermal Drilling Cost Curve Updates," 2025) [24], as do open-source techno-economic tools — the System Advisor Model (SAM) for site-based capex, O&M and LCOE estimation, and GeoPHIRES for dynamic reservoir performance and heat extraction over time [28]. National-lab techno-economic analysis has been used to inform "electricity grid capacity expansion models, and next-generation technology cost targets" [28] — i.e., the machinery for scenario-building exists, but the resulting scenario numbers are not in evidence here.

The **assumed mechanism** of learning is, however, clearly stated: technology transfer from oil and gas. Recent advances involve "applying oil and gas developed techniques such as long horizontal laterals and multi-stage hydraulic fracturing to geothermal settings" [1][15], and Fervo Energy is credited with a milestone "in using oil drilling technology to tap geothermal energy" [20]. Implicitly, roadmaps that assume rapid cost decline are borrowing the shale learning curve rather than deriving a geothermal-specific one — but the corpus does not quantify that borrowing.

Milestones that would validate the scenarios

Already claimed as achieved:

1. *Fracture network creation plus sustained circulation in hot impermeable rock.* Recent and ongoing demonstrations — Fervo, FORGE, and DOE EGS demonstrations — "show successful creation of a fracture network in deep and hot impermeable rocks combined with long-term fluid circulation" [1][15]. This is the foundational technical milestone. 2. *Commercial-scale first-of-a-kind demonstration*, reported for Fervo in 2023 [20]. 3. *Heat extraction at reasonable rates from hydraulically stimulated low-permeability hot crystalline rock*, established at Fenton Hill, with production temperature rising to ~190 °C at ~10 MW thermal over a 30-day test [14][15]. 4. *Multi-zone and multi-producer well architectures*, tested at Soultz [18].

Not yet demonstrated, and therefore the live validation tests:

1. *Scale jump from tens of MW to GW.* The largest EGS project in evidence remains a 25 MW demonstration [8]; construction of what could become "the world's largest enhanced geothermal plant" in Utah is referenced but not quantified [20]. A confirmed multi-hundred-MW plant would be the first genuine test of the 100 GWe pathway. 6. *Cost verification against $45/MWh by 2035* [23], measured against ATB capex, well productivity and O&M assumptions [24]. 7. *Long-term thermal decline within the assumed 20–30 year economic lifetime* [8]. Reservoir modelling work on thermal drawdown, well layout, fracture parameters, and water-cooling effects [10][12][15] indicates this is treated as an open question, not a settled one. 8. *Drilling to depth and temperature at cost.* Advanced drilling is said to be able to penetrate hard crystalline rock to depths "up to, or exceeding, 15 km," accessing rock at 400 °C and above [8]; superhot rock resources with up to ten times the output of a normal geothermal well remain unharnessed "due to significant technical and technological challenges" [30]. Delivering superhot-rock wells at commercial cost would be a step-change validation; continued failure to do so caps the upside. 9. *Sustained policy and funding continuity.* The Infrastructure Investment and Jobs Act authorized $84 million for EGS development through four demonstration projects, and the Inflation Reduction Act extended the production tax credit for geothermal and included it in the new Clean Electricity PTC from 2024 [32]. Earlier cycles offered up to $84 million over six years in 2009 FOAs and $350 million in Recovery Act funding including $80 million specifically for EGS [33][35]. Completion of the four IIJA demonstrations is a discrete, checkable milestone.

Events that would falsify or derail the scenarios

- **Induced seismicity.** "Seismicity is common in EGS, because of the high pressures involved," and events at The Geysers correlate with injection activity [34]. Induced seismicity led Basel to suspend and then cancel its project [34][18], and the 2017 M5.4 Pohang earthquake "may have been linked to the activity of the Pohang EGS project," after which all research activities stopped in 2018 [22][34]. These are the clearest historical instances of a scenario being falsified by non-technical failure. **The corpus contains a contrary view:** the Australian government holds that hydrofracturing-induced seismicity risks "are low compared to that of natural earthquakes, and can be reduced by careful management and monitoring" and "should not be regarded as an impediment to further development," while noting seismicity varies by site and must be assessed before large-scale injection [34]. Mitigation is being institutionalised — a traffic-light system for deep geothermal well stimulation was designed and implemented in Finland [6] — so a validating milestone would be repeated large-volume stimulations under such protocols without a felt event.

- **Systematic site-level underperformance.** A survey titled "What Are the Challenges in Developing Enhanced Geothermal Systems (EGS)? Observations from 64 EGS Sites" [27] indicates a substantial record of difficulty across the historical project population, which spans 1970 to the 2010s across Iceland, the U.S., Germany, France, Japan, Australia and elsewhere [2][3].
- **Funding fragility.** The original Fenton Hill programme was terminated by budget cuts, not by technical failure [14] — a precedent for roadmaps collapsing on appropriations rather than physics.
- **Supply-chain and workforce limits.** The corpus asserts that deployment at tens of gigawatts "would face constraints in skilled crews and high-temperature tools" [20]; however, the supporting text for that entry is a bibliography listing rather than substantive analysis, so this should be treated as an assertion needing better sourcing.
- **Competition from non-stimulated alternatives.** Closed-loop geothermal requires no reservoir stimulation, "which limits the risk of induced seismicity and lowers water consumption," can theoretically be applied anywhere, and offers heat-production estimates with relatively high confidence — though it faces complex downhole completions and heat-transfer-area challenges [29]. If closed-loop or superhot-rock pathways scale faster, EGS-specific gigawatt roadmaps could be superseded rather than falsified. Comparative techno-economic work suggests EGS "can be competitive with closed-loop/advanced geothermal (AGS) and conventional hydrothermal systems under specific conditions" [15][27] — a conditional, not unconditional, verdict.

Summary judgement on the state of the roadmaps

The corpus supports the following characterisation: there is one long-standing capacity scenario (100 GWe U.S. by 2050, on $1B/15yr R&D, against a 1.2–12.2 TW recoverable base) [17], one aggressive cost target (90% reduction to $45/MWh by 2035) [23], a demonstrated technical foundation (fracture networks plus long-term circulation) [1], and a deployed capacity base still measured in single- and double-digit megawatts [8][18]. The critical missing links — in the evidence and arguably in the roadmaps themselves — are (a) an explicit learning rate connecting demonstration cost to $45/MWh, and (b) multi-year field data on thermal decline and well productivity, which is precisely the parameter set that capacity-expansion models depend on [24]. Until those are published, the roadmaps are best treated as targets with identified levers rather than as calibrated forecasts.

Conclusions

What the evidence establishes

Enhanced geothermal systems (EGS) are best understood as an engineering answer to a resource-access problem: most geothermal heat reachable by conventional techniques sits in dry, impermeable rock, and EGS uses stimulation — chiefly hydraulic — to make that rock produce [4]. Permeability is created or restored either by hydro-shearing, in which high-pressure injection raises fluid pressure and triggers shear that dilates pre-existing fractures without requiring proppants [8], or by tensile hydraulic fracturing of the kind used in oil and gas, which can create wholly new fractures [8]. In practice the field has used a broader toolkit than hydraulics alone: hydraulic, chemical, thermal, carbon-based, and explosive stimulation have all been applied, sometimes in combination, and some projects sit at the margins of existing hydrothermal fields where wells intersect hot but tight rock [2][7]. Documented variants include cyclic waterfracturing [11], massive hydraulic fracturing in low-permeability sediments [13], acid and chemical treatments [5], and radial jetting [6]; chemical stimulation is being tested at Utah FORGE as an alternative or complement to mechanical methods [1].

The technology has a long and geographically broad experimental record. Fenton Hill, New Mexico (from 1973, reservoir completed 1977 at ~2.6 km and 185 °C) was the first attempt at a deep full-scale EGS reservoir and showed that heat could be extracted at reasonable rates from hydraulically stimulated low-permeability crystalline rock, reaching ~10 MW thermal in a 30-day flow test before budget cuts ended the work [14]. Dozens of subsequent projects across Iceland, Germany, France, the UK, Japan, Australia, the Philippines, the US and elsewhere are catalogued with their start years and stimulation methods [2]. Soultz-sous-Forêts connected a 1.5 MW demonstration plant to the grid and tested multi-zone stimulation and triplet (one injector/ two producer) well configurations [18]. The most recent phase is defined by transfer of unconventional oil-and-gas practice — long horizontal laterals and multi-stage hydraulic fracturing — into geothermal settings, with Fervo, Utah FORGE and DOE demonstrations reported to have created fracture networks in deep, hot, impermeable rock and sustained long-term circulation [1][20]. Complementary design questions (multi-well injection, well layout, downhole flow control, spiral wellbores, fracture-network complexity) are being explored largely through numerical simulation rather than field data [10][12][15][16][25], and national-lab work adds reservoir modelling, high-performance computing, slender-body optimisation models and techno-economic tools such as SAM and GEOPHIRES [19][21][28].

Potential, cost and policy

The anchor resource assessment remains the 2006 MIT study: >13,000 ZJ of US EGS resource at 3–10 km, >200 ZJ extractable (potentially >2,000 ZJ with better technology), 1.2–12.2 TW “recoverable” with then-current technology, and 100 GWe or more deployable by 2050 on a $1 billion, 15-year R&D investment [17]. Site-level potential can be large: Cooper Basin alone is credited with 5,000–10,000 MW [8], and EGS is described as feasible essentially anywhere given sufficient depth, with plant lifetimes of 20–30 years and baseload output [8]. Deeper, hotter targets — superhot rock above ~400 °C — could yield up to ten times the output of a normal geothermal well but have not yet been harnessed for power because of unresolved technical and technological challenges [30].

On cost, the corpus contains an unresolved tension. MIT (2006) claimed EGS electricity as low as 3.9 cents/ kWh, sensitive to resource temperature, flow rate, drilling cost and conversion efficiency [17]. Yet in 2022 DOE’s Enhanced Geothermal Shot set a target of cutting EGS cost by 90% to $45/MWh (4.5 cents/kWh) by 2035 [23], which implies a present-day cost roughly an order of magnitude above the MIT figure. The corpus does not reconcile these numbers; current levelized-cost values and their underlying capex, capacity-factor, well-productivity and decline assumptions are said to reside in the Electricity Annual Technology Baseline [24], but the specific figures are not in evidence here. Public support is documented — $84 million in DOE funding opportunities in 2009 plus $350 million of ARRA funds including $80 million for EGS [33], $84 million under the Infrastructure Investment and Jobs Act for four demonstrations, and PTC/Clean Electricity PTC eligibility under the Inflation Reduction Act [32], with a £15 million award for UK deep geothermal [35] — but the corpus offers no systematic picture of private capital flows, project finance structures, or cost-of-capital.

Risks and conflicting evidence

Induced seismicity is the principal documented risk and is described as common in EGS because of the pressures involved, with events at The Geysers correlated to injection [34]. Basel’s project was suspended and then cancelled after induced seismicity [34][18], and the 2017 Mw 5.4 Pohang earthquake may have been linked to the Pohang EGS project, which halted all research in 2018 [22][6]. Against this, the Australian government’s position cited in the corpus is that hydrofracturing-induced seismicity risk is low relative to natural earthquakes, manageable by monitoring, and should not impede development [34]. Both positions are in the record; the corpus does not adjudicate between them, though it notes site-specific assessment is needed before large-scale injection [34] and documents mitigation practice such as the traffic-light system implemented in Finland [6]. A structural contrast is that closed-loop geothermal requires no stimulation, which limits seismicity risk and water use, at the cost of complex downhole completions and limited heat-transfer area [29].

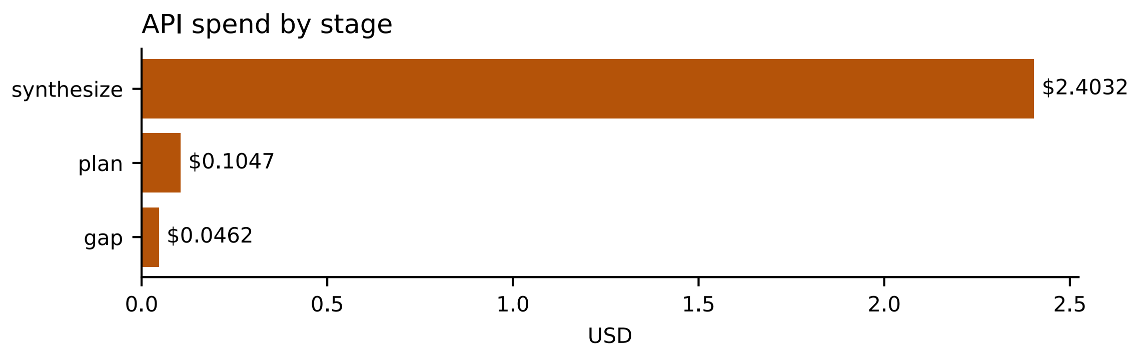
A second, minor conflict: one source states the world's largest EGS project is the 25 MW Cooper Basin demonstration in Australia [8], while another cites reporting that Utah could become home to the world's largest enhanced geothermal plant [20]. These are plausibly statements from different dates rather than a substantive disagreement, but the corpus does not settle current rankings.

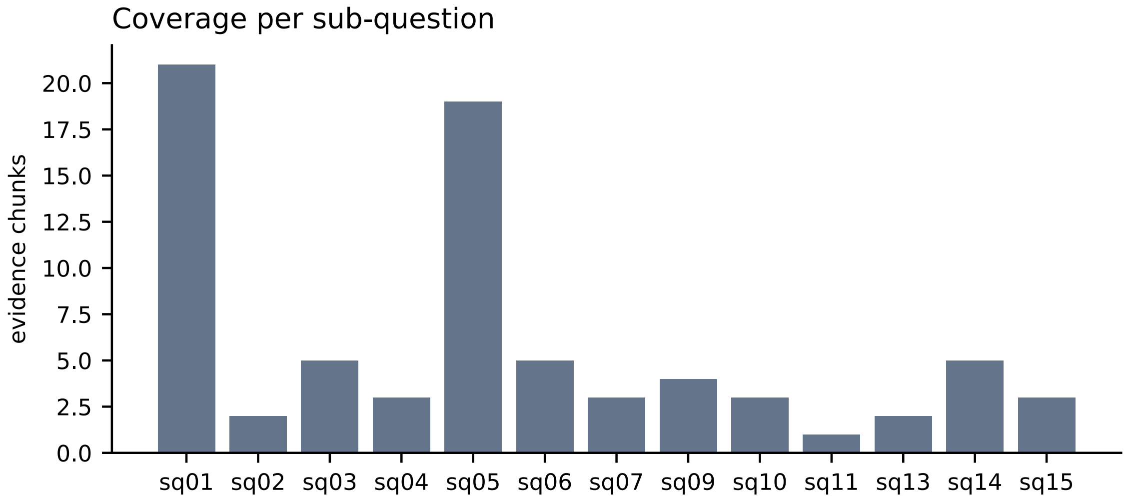
Open questions and evidence gaps

- **Measured field performance.** Beyond Fenton Hill's ~10 MW thermal test [14] and Soultz's 1.5 MW plant [18], the corpus provides no flow rates, production temperatures, decline rates or capacity factors for the recent Fervo/FORGE demonstrations; the claim of successful long-term circulation is asserted at a summary level [1][20].
- **Drilling costs.** The transfer of oil-and-gas drilling technology is documented [1][20] and a drilling cost-curve update exists [24], but no quantitative cost-per-well or learning-rate data appear in the corpus.
- **Scale-up constraints.** One entry asserts that gigawatt-scale deployment would be limited by skilled crews and high-temperature tools, but the supporting material is a bibliographic title rather than substantive data [20]; workforce, supply-chain and high-temperature-tool constraints are therefore essentially unevidenced here.
- **Water use, land use, lifecycle emissions and permitting** for EGS are not covered by the corpus, other than the comparative note that closed-loop avoids stimulation-related water consumption [29].
- **Non-US resource assessments** are absent; the quantitative potential estimates available are US-specific and nearly two decades old [17], with newer national-scale mapping and temperature-at-depth/EGS potential characterisation described but not quantified [31][26].
- Several claims rest only on reference-list titles from a single encyclopedia article [3][5][6][9][11][13][20][35]; these should be treated as pointers to primary literature rather than as independent findings. The same source set also uses "NREL" and "NLR" interchangeably for the same laboratory [1][19], which introduces minor ambiguity about provenance.

On balance, the evidence supports a conclusion that EGS has moved from repeated small-scale experimentation [2][14][18] to credible field demonstration of engineered fracture networks and circulation [1][20], with a very large but coarsely characterised resource [17][8], and that the decisive open issues are cost trajectory — where the corpus's own figures disagree by roughly an order of magnitude [17][23] — and the site-specific management of induced seismicity, on which authoritative sources differ in emphasis [34][22].

Run Charts

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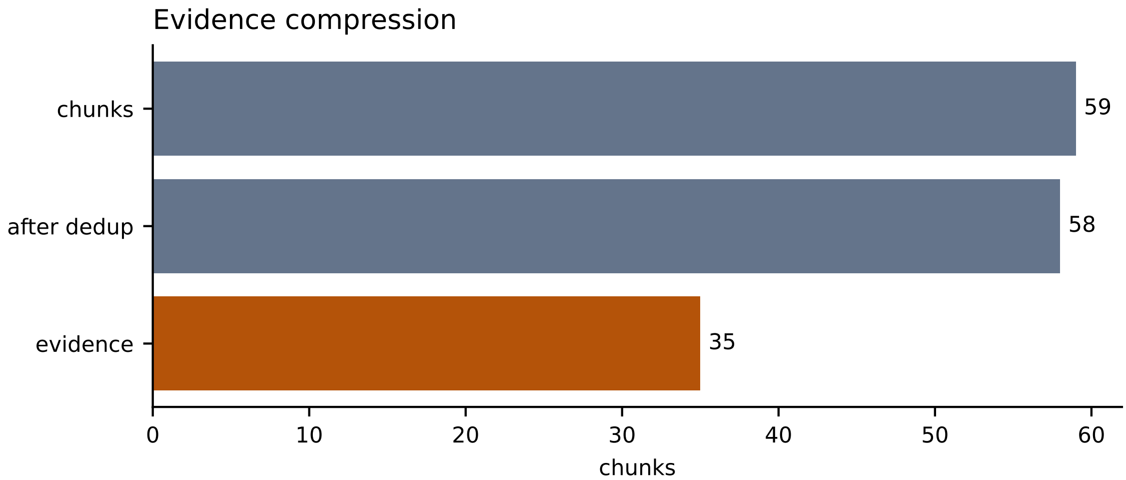
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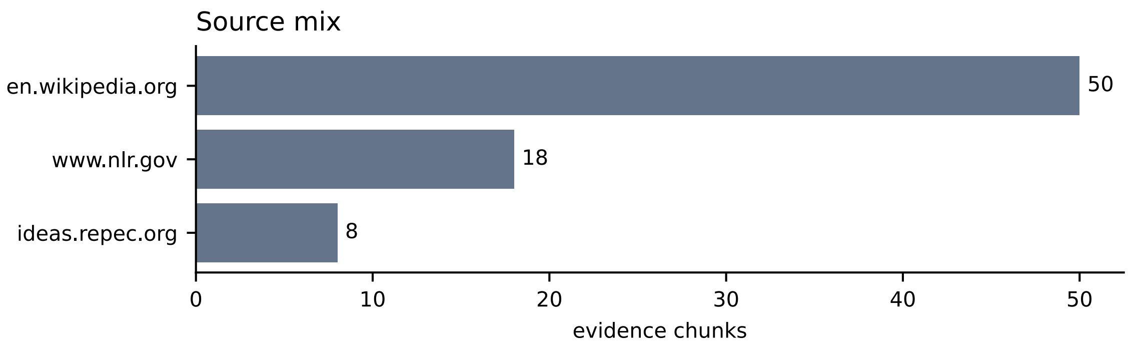
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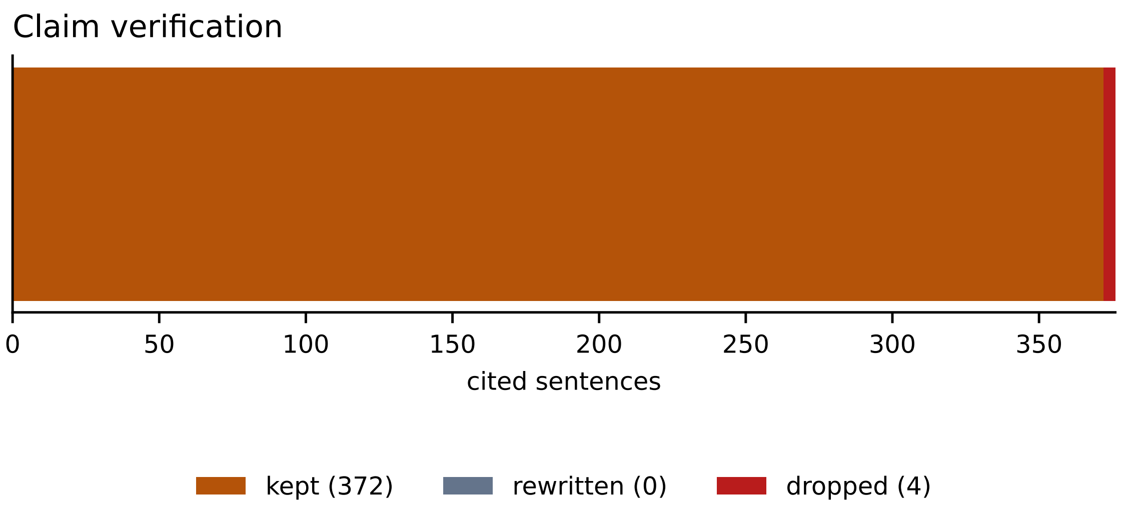

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Run Facts

Run id	6bc4d96f41d3
Engine	premium
Iterations	2
Sources fetched	10
Chunks	59
After dedup	58
Evidence chunks	35
Claims checked	376
Claims rewritten / dropped	0 / 4
API spend	USD 2.5540
Spend: synthesize	USD 2.4032
Spend: plan	USD 0.1047
Spend: gap	USD 0.0462